Check for updates

Water Science & Technology

IWA PUBLISHING

© 2026 The Authors

Water Science & Technology Vol 93 No 6, 817 doi: 10.2166/wst.2026.234

Applying machine learning and AI for nanofiltration membranes water applications: a review

Azzam Abuhabib [a,*], Ahmed Albahnasavi[b], Heba Isawi[c,d], Law Yong Ng[e], Abdul Wahab Mohammad [f] and Sofiah Hamzah[g]

[a] Department of Water & Environmental Engineering, Faculty of Civil Engineering, Universiti Teknologi Malaysia UTM, Johor Bahru, Malaysia
[b] Environmental Engineering Department, Gebze Technical University, Gebze, Turkey
[c] Water Treatment & Desalination Unit, Hydrogeochemistry Department, Water Resources & Desert Soils Division, Desert Research Center, Cairo 11753, Egypt
[d] Egypt Desalination Research Center of Excellence, Desert Research Center, Cairo 11753, Egypt
[e] Univerisit Tunku Abdul Rahman (UTAR), Johor Bahru, Malaysia
[f] Chemical and Water Desalination, Engineering Program, University of Sharjah, Sharjah, United Arab Emirates (UAE)
[g] Faculty of Ocean Engineering, Universiti Malaysia Terengganu (UMT), Terengganu, Malaysia
*Corresponding author. E-mail: amazzam@utm.my

AA, 0000-0002-1015-5933

ABSTRACT

Nanofiltration (NF) has long been the focus of researchers and operators in the field of water and wastewater treatment and desalination. Characterised by both high rejection of diverse substances depending on their molecular composition and charge, and high flux, NF membranes have demonstrated broad applicability while maintaining long-lasting performance. However, NF membrane scalability and reliability are intrinsically constrained by non-linear phenomena like complex solute-rejection mechanisms, long-term fouling dynamics, and the inherent flux–selectivity trade-off. Traditional mechanistic models relying on simplifying assumptions within the solution–diffusion framework, often fail to accurately predict performance in heterogeneous, real-time water matrices, leading to a critical divergence between predictive capability and operational reality. This review addresses this gap by comprehensively assessing the integration of machine learning (ML) and artificial intelligence (AI) across three critical domains: fabrication optimisation, performance prediction, and fouling diagnosis and mitigation, through the analysis of 100 NF-ML publications over the past 10 years. Additionally, it highlights key methodological challenges including data scarcity and heterogeneity, and lack of robust external validation. Finally, the review emphasises that future advancements lie in reinforcement learning for real-time adaptive control and in hybrid ML-mechanistic frameworks bridging the existing gap between data-driven prediction and transparent mechanistic understanding.

Key words: machine learning, membrane, nanofiltration, water applications

HIGHLIGHTS

- Illustration of machine learning (ML) approaches and algorithms applicable to nanofiltration (NF) membranes.
- Explanation of inputs, outputs, and processes in between, all of which act as a roadmap on how ML flow looks for NF membrane processes and applications.
- Illustration of challenges and solutions for dataset availability and processing, and ML models' accuracy and interpretability.

1. INTRODUCTION

Driven by population growth and environmental stress, the need for reliable freshwater is currently exceeding global supply capacities. Projections estimate that two-thirds of the world's population will soon face water insecurity. However, these figures reveal only a fraction of the broader systematic challenges in water management.

However, these figures alone reveal only part of the challenge. More critically, they underscore the inadequacy of current treatment strategies and the urgent need for sustainable, high-efficiency purification technologies capable of addressing complex and variable water quality (Isawi 2023; El-Aassar *et al.* 2024). In this context, membrane-based desalination and water treatment have transitioned from supplementary options to central components of global water security frameworks. Beyond

This is an Open Access article distributed under the terms of the Creative Commons Attribution Licence (CC BY 4.0), which permits copying, adaptation and redistribution, provided the original work is properly cited (http://creativecommons.org/licenses/by/4.0/).

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

the well-established benefits of modularity and selectivity, these systems are gaining traction due to their ability to outperform traditional treatment processes. Specifically, they offer enhanced energy efficiency and a reduced environmental footprint while maintaining adaptability to complex feed compositions. However, despite their widespread deployment, significant performance trade-offs remain across membrane classes, especially regarding fouling behaviour, selectivity/permeability limitations, and cost-efficiency balances (Isawi 2018, 2023). Several membrane-based water treatment methods, such as ultrafiltration (UF) (Isawi 2019), nanofiltration (NF) (Di *et al.* 2024), forward osmosis (El-Noss *et al.* 2020; Ahmed *et al.* 2021), and reverse osmosis (RO) (Isawi *et al.* 2016; Qrenawi & Abuhabib 2020), as well as distillation, and electrochemical processes (Shahlol *et al.* 2020; Isawi 2024), are widely applied to remove undesirable substances and produce freshwater (Figure 1). Among these, as a pressure-driven process characterised by a semi-porous structure, NF serves as an intermediary technology bridging the gap between UF and RO. These membranes typically feature effective pore diameters in the sub-nanometer range and a molecular weight cut-off (MWCO) of 0.2–10 kDa (Behroozi & Ataabadi 2021; Meng *et al.* 2023; Usha *et al.* 2025). Due to this intermediate selectivity, NF operates at significantly lower operational pressure than RO while achieving higher permeation flux. Consequently, NF has emerged as a highly attractive solution for diverse water applications including brackish water desalination, wastewater polishing, and seawater pre-treatment. Having too much in common in terms properties and selectivity as a pressure-driven process, both RO and NF have high similarity when it comes to water applications. Practically, NF membranes are highly regarded for their robust capacity to selectively separate inorganic salts and low MW organic molecules from aqueous solutions. These systems occupy a distinct operational niche, offering separation efficiencies and transport characteristics that intermediate between the coarser UF and the more-dense RO processes. Unlike RO, NF membranes reject monovalent ions at a lower rate, while effectively rejecting divalent ions at a higher rate and maintaining high

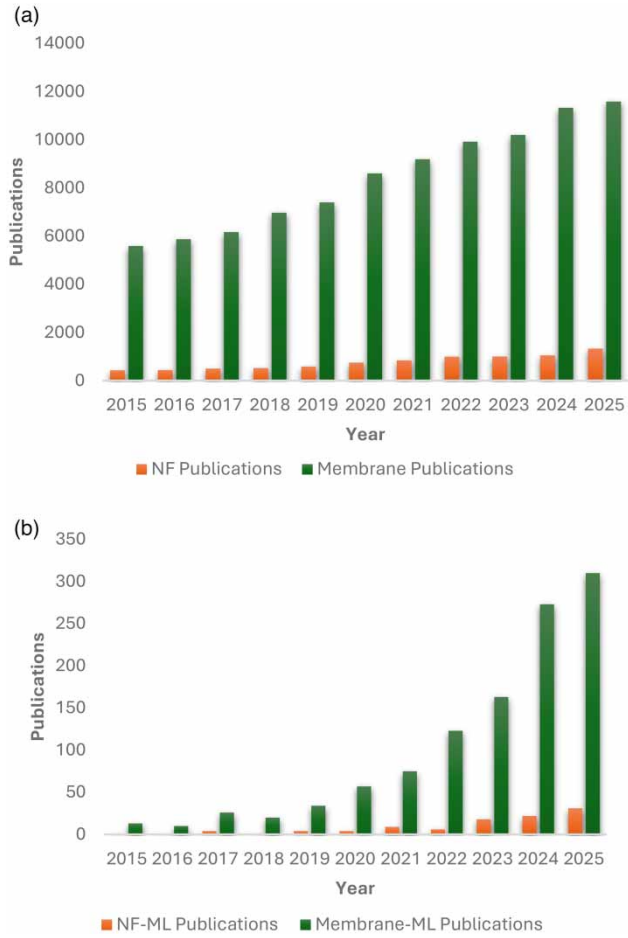


Figure 1 | Publications per year for (a) NF membranes vs. membrane and (b) NF-ML vs. membrane-ML – accessed on November 2025 (prepared by the researchers).

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

flux. Despite these advantages, NF membranes face critical challenges that limit their broader application, including membrane fouling, limited robustness under harsh chemical environments, and the classical trade-off between permeability and selectivity. Addressing these issues requires not only material innovations but also a deeper understanding of transport mechanisms, surface-solute interactions, and system-level optimisation. Therefore, a comprehensive and critical assessment of NF membranes development, performance, trends, and application opportunities is essential to guide next generation of research and industrial adoption. Table 1 illustrates the number of commercial NF membranes available in the market along with their manufacturers and important features.

Although NF research has a long history extending back several decades, the analysis in this review intentionally focuses on the most recent 10-year period, during which the acceleration of NF research has been particularly. The past 10 years witnessed growing interest in exploring and investigating NF membranes applications as indicated by the significant number of publications found in Scopus database (Figure 1) where the number of publications increased from 432 in 2015 to 1,332 in 2025. Out of membrane water-based publications (5,592 in 2015 and 11,587 in 2025), NF membranes witnessed an increase in percentage of publications from 7.7% in 2015 to 11.5% in 2025. When machine learning (ML) included in the search for both NF membranes and membranes in general for water-based applications, both witnessed significant increase over the same period with NF-ML-based publications represents 10% out of total membrane-ML-based publications. NF membranes have been increasingly applied in complex water and wastewater matrices, where performance is governed by a non-linear interplay of membrane structural properties, feed composition, operating conditions, and fouling dynamics (Boussouga *et al.* 2022). These systems are often difficult to describe adequately with purely mechanistic models, especially when many variables interact simultaneously and when data are heterogeneous or scarce (Du *et al.* 2022). The rapid expansion of ML and broader artificial intelligence (AI) frameworks has provided powerful data-driven methodologies for capturing complex non-linear relationships within membrane systems. These computational tools effectively support both the predictive modelling of key performance indicators, such as flux, rejection, and fouling indices, and the inverse design of membrane structures and optimal operating conditions (Li *et al.* 2023). Recent reviews have discussed AI and ML in membrane technology at a general level, typically combining UF, NF, RO, and organic solvent nanofiltration (OSN) within a single framework.

This review aims to comprehensively examine the advancements and progress in applying ML to NF membrane research. This is supported by a particular focus on key thematic areas such as predictive modelling of membrane fabrication and surface modification, performance optimisation through data-driven characterisation, and ML-enabled solutions to address membrane fouling. It further explores the role of ML in expanding industrial and environmental applications of NF membranes. Such exploration is done by identifying emerging trends, knowledge gaps, available models and algorithms and their applicability, and future research directions that could accelerate the integration of ML in NF membrane innovation and promote their wider adoption across sectors.

2. SEARCHING INCLUSION/EXCLUSION CRITERIA AND MAJOR CONTRIBUTION OF THIS REVIEW

The major research database used for screening was Web of Science and Scopus databases, giving their reliability and comprehensiveness. The inclusion criteria were exclusively dependent on selected keywords connecting NF membranes to water desalination and wastewater applications, both of which were integrated with ML. As such, the specific keywords were used accordingly: NF, membrane, water desalination, wastewater treatment, ML, ANN, deep learning (DL), modelling.

This review provides several distinguished benefits and contributions, including:

- Illustration of ML approaches and algorithms applicable for NF membranes linked to three major fields: fabrication and modification, operational performance, and fouling mechanisms and control. For each field, several concrete examples from different publications were illustrated showing commonly used algorithms along with their level of accuracy and results obtained.
- Explanation of inputs, outputs, and processes in between, all of which act as a roadmap on how ML flow looks for NF membrane processes and applications. Additionally, levels and categories of algorithms applied were illustrated (e.g., supervised/unsupervised, ensembled/reinforced learning).
- Illustration of challenges and solutions for dataset availability and processing, and ML models' accuracy and interpretability.

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

Table 1 | Commercial NF membranes in the market with their related features (prepared by the researchers)

Membrane	Manufacturer	MWCO (Da)	Maximum temperature (°C)	pH range	Stabilised salt rejection (%)	Composition on the top layer
NF270	Dow Filmtec	200–400	45	2–11	>97%	Polyamide thin-film composite
NF200	Dow Filmtec	200–400	45	3–10	50–65% CaC_{l2}, 3% $MgSO_4$, 5% Altrazine	Polyamide thin-film composite
NF90	Dow Filmtec	200–400	45	3–10	85–95% NaCl	Polyamide thin-film composite
TS80	TriSep	150	45	2–11	>99%	Polyamide
TS40	TriSep	200	50	3–10	>99%	Polypiperazineamide
XN45	TriSep	500	45	2–11	95%	Polyamide
UTC20	Toray	500	35	3–10	60%	Polypiperazineamide
TR60	Toray	400	35	3–8	55%	Cross-linked polyamide composite
CK	GE Osmonics	400	30	5–6.5	94% $MgSO_4$	Cellulose acetate
DK	GE Osmonics	150–300	50	3–9	98% $MgSO_4$	Cross-linked aromatic polyamide
DL	GE Osmonics	150–300	90	6–9	96% $MgSO_4$	Cross-linked aromatic polyamide
HL	GE Osmonics	150–300	50	3–9	98% $MgSO_4$	Cross-linked aromatic polyamide
NFW	Synder	300–500	50	3–10.5	97% $MgSO_4$, 40% NaCl	Proprietary polyamide thin-film composite
NFG	Synder	600–800	50	4–10	50% $MgSO_4$, 10% NaCl	Proprietary polyamide thin-film composite
TFC SR100	Koch	200	50	4–10	>99%	Proprietary thin-film composite polyamide
SR3D	Koch	200	50	4–10	>99%	Proprietary thin-film composite polyamide
SPIRAPRO	Koch	200	50	3–10	99%	Proprietary thin-film composite polyamide
ESNA1	Nitto-Denko	100–300	45	2–14	89%	Composite polyamide
NTR7450	Nitto-Denko	600–800	40	2–14	50%	Sulfonated polyethersulfone
ESNA1-LF-LD	Hydranautics	100	45	2–10	93% $CaCl_2$	Composite polyamide
ESNA1-LF2-LD	Hydranautics	100	45	2–10	91% $CaCl_2$	Composite polyamide
ESNA4-LD	Hydranautics	120	45	2–10	>99% $MgSO_4$	Composite polyamide
NANO-SW-4040	Hydranautics	120	45	3–9	99.8% $MgSO_4$	Composite polyamide
NANO-SW	Hydranautics	120	45	3–9	99.8% $MgSO_4$	Composite polyamide
NANO-SW MAX	Hydranautics	120	45	3–9	99.8% $MgSO_4$	Composite polyamide
HYDRACoRe10-LD-4040	Hydranautics	3,000	45	2–11	5–30% NaCl	Sulfonated polyethersulfone
HYDRACoRe50-LD-4040	Hydranautics	1,000	45	2–11	40–65% NaCl	Sulfonated polyethersulfone
NF	Alfa Laval	120	5–50	3–10	≥99% $MgSO_4$	Polyester
NF99HF	Alfa Laval	120	5–50	3–10	≥99% $MgSO_4$	Polyester
NFS	Synder	100–250	50	2–11	99.5% $MgSO_4$, 50% NaCl	Proprietary thin-film polyamide
NFX	Synder	150–300	50	2–11	99% $MgSO_4$, 40% NaCl	Proprietary thin-film polyamide
NFW	Synder	300–500	50	3–10	97% $MgSO_4$, 20% NaCl	Proprietary thin-film polyamide
NFG	Synder	600–800	50	3–10	50% $MgSO_4$, 10% NaCl	Proprietary thin-film polyamide

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

- Valuable comparison between ML models and mechanistic and mathematical ones, pointing out the advantages and disadvantages of each of them.

3. OVERVIEW OF NF MEMBRANE TECHNOLOGY

3.1. NF membrane separation fundamentals and applications

NF membrane is primarily a separation process in which an aqueous solution is driven through by applied pressure. Certain substances are rejected at different rates, influenced by a variety of factors including the geometry and charge of the membrane surface, as well as the ionic composition and charge of the solution. NF membranes are crafted with a specific MWCO that permits the retention of certain solutes while allowing others to flow through. In addition, the significant influence of the NF membranes filtration process by hydrophobic and hydrophilic forces interactions contributes to solute size exclusion and the Donan effect and attributing to enabling the targeted removal of divalent ions, small organic compounds, and viruses from water solutions (Donnan 1995; Li *et al.* 2022). Practically, solute transport across the NF membrane occurs through three primary mechanisms: convection, diffusion, and electromigration. Convection occurs in a thick, highly porous membrane with high permeate flux. Diffusion takes place in a thin membrane having a high surface charge density with low porosity and permeate flux. Electromigration is directly linked to a moderate charge density at the membrane surface while a solute passes through. Furthermore, hydrophobic and hydrophilic interactions mechanisms are influential to the uncharged solutes adsorption and rejection depending on these solutes hydrophobic or hydrophilic properties. For example, greater adsorption on hydrophobic membrane surfaces occurs due to hydrophobic interactions, leading to have higher rejection rates (Wang *et al.* 2021, 2024e), while surface charge and pore size influence hydrophilic interactions depending on the solute size and polarity (Saeid Rostami & Mehdi Khodaei 2023; Wang *et al.* 2024e). These key performance features and mechanisms have earned NF membranes wide applicability in many water industries. They have been widely used and applied in various industries, most of which water applications including water desalination and treatment (Voutchkov 2018; Belhamidi *et al.* 2022; Mallya *et al.* 2023; El-Ghzizel *et al.* 2024; Idrissi *et al.* 2024; Laurell *et al.* 2024), wastewater treatment (Samavati *et al.* 2023, 2024; Al Kholif *et al.* 2024), and food industry (Abdorrezaee & Raisi 2021; Roselli *et al.* 2025). The major focus on NF membranes application is commonly the water applications with some interest on other non-water applications whereby NF membranes are applied in the form of OSN (Alqadhi *et al.* 2024; Waheed *et al.* 2024; Roselli *et al.* 2025).

3.2. Traditional challenges in NF membrane design and operation

NF membrane design and operation, as well as performance modelling, remain at the focus of membrane manufacturers and researchers. Consequently, at the heart of such focus is having an NF membrane that fits certain uses with desirable features and sustainably maintained performance with respect to an acceptable operational condition (Zahid *et al.* 2018; Usha *et al.* 2025). On the other hand, fouling continues to rank as the number one challenge in membrane technology in general, and NF membrane is no exception. It can also be seen as the performance turning point where deterioration is observed, and symptoms begin to surface, obviously indicated by a decline in flux (recovery reduction). This is where energy consumption starts to increase, and maintenance and cleaning become unusually more frequent. It is hard to tell when the fouling would occur, but mostly, it will take some time until it becomes noticeable, as the membrane performance or basically flux declining may occur gradually or at a very minor magnitude. Despite the massive number of studies and research conducted to understand membrane fouling and the way to control or mitigate it, the need to further understand and explore the phenomena of how it happens and parameters affecting or contributing to it, as well as controlling measures are still vivid and growing (Al Kholif *et al.* 2024; Borgomeo 2024; Liu *et al.* 2024). Fundamentally, concentration polarisation and osmotic pressure are potentially the phenomenal mechanisms involved in fouling. In concentration polarisation, the retained solutes are accumulated in the membrane boundary on the feed side leading to form high solute concentration on the membrane surface. On the other hand, the osmatic pressure increasing because of accumulated solutes (organic or inorganic), which will consequently lead to the deterioration of the transmembrane pressure and permeate flux (Mohammad *et al.* 2015). In general, NF membrane fouling can be categorised based on the type that is linked to the foulant species causing the fouling. Practically, there are several major factors affecting or influencing NF membranes fouling that can be summarised as (Zheng *et al.* 2024b):

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

- Raw water/feed characteristics (natural organic matter (NOM) concentration, ionic interaction, pH, and temperature).
- Membrane characteristics (morphology, structure, hydrophilicity, and other surface properties).
- Operational conditions (pressure, recovery rate, and hydrodynamic conditions).

Given the fact that membrane fouling is inevitable, mitigation and control are essential to reduce the negative effects caused by fouling to maintain membrane performance and reserve membrane lifespan. Based on the illustration made above for factors influencing membrane fouling, mitigation measures can be suggested to deal with these factors accordingly (Liu *et al.* 2024; Zheng *et al.* 2024b). For instance, controlling and optimising feed water and parameters operational conditions, as well as pre-treatment methods, could be the key to reducing fouling effectively. In addition, the use of some materials and additives that can increase membrane fouling resistance in fabrication or surface modification processes may also reduce fouling tendency as preventive measures.

3.3. NF membrane modelling

NF membranes modelling plays a crucial role in deepening theoretical understanding, guiding effective maintenance strategies, and driving performance improvements (Mohammad *et al.* 2015; Al Kholif *et al.* 2024; Petukhov & Johnson 2024; Xu *et al.* 2024). Previously, the approach has evolved from empirical black-box models rooted in irreversible thermodynamics to models based on the extended Nernst–Planck equation (Tsuru *et al.* 1991; Levenstein *et al.* 1996). The latter has gained prominence due to its ability to provide insights into the properties of both the membrane and the process stream. Such models aim at integrating as much physical realism as possible into the membrane processes representation to enable a closer alignment between measurable quantities and adjustable model parameters. However, reliance on pore radius and membrane charge on characteristic parameters significantly limited NF modelling, as they are difficult to measure at the near-atomic length scales. The highly complex neutral solutes and charged electrolytes are deeply governed by nanoscale phenomena, posting a rigorous challenge to any macroscopic continuum model of ion transport and partitioning. Similarly, molecular dynamics simulations with more precise physical descriptions are constrained by the physical structure understanding which is highly limited to real NF membranes electrical properties and streams. Consequently, advancements in NF modelling have progressed alongside improvements in measurement techniques for characterising NF membranes and process streams. These enhanced techniques are critical for validating the appropriateness of model parameters, ensuring that the models can effectively represent the intricate phenomena occurring within NF systems. Ion transportation is mathematically described by the extended equation of Nernst–Planck (Equation (1)), while Equation (2) is the full extended Nernst–Planck that can be stated as the ion transportation in a solution governing equation, practically describing ion transportation into membrane pore similar to the one observed for NF and RO membranes (Giddings *et al.* 1968; Deen 1987; Donnan 1995).

$$J_i = -\frac{c_i K_{i,d} D_{i,\infty}}{RT}\frac{d\mu}{dx} + K_{i,c} c_i V \tag{1}$$

$$j_i = -D_{i,p}\frac{dc_i}{dx} - \frac{c_i D_{i,p}}{\gamma_i}\frac{d\gamma_i}{dx} - \frac{c_i D_{i,p}}{RT} V_{si}\frac{dP}{dx} - \frac{c_i D_{i,p}}{RT} z_i F \frac{d\varphi}{dx} + K_{i,c} c_i V \tag{2}$$

where j_i is the ionic flux, c is the concentration, V is the solvent velocity, and $K_{i,c}$ and $K_{i,d}$ are hindrance factors to account for the convection and diffusion inside a confined space.

4. AI, ML, AND DL: AN OVERVIEW

4.1. AI and ML concepts and algorithms

ML can be broadly understood as a data-driven framework in which computational models learn relationships between inputs and outputs by optimising performance through experience, rather than relying solely on predefined physical equations (Mitchell 1997). In the context of NF research, ML is not employed to replace fundamental transport theory, but rather to complement it by capturing complex, non-linear interactions between membrane properties, operating conditions, and performance indicators such as flux, rejection, and fouling behaviour.

Unlike conventional hypothesis-driven modelling approaches, where system behaviour is inferred from simplified physical assumptions, ML enables direct learning from experimental or operational data when sufficient volume and quality are available (Bishop & Nasrabadi 2006; Madhavan 2025). This paradigm shift has been accelerated by advances in computational

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

power and the increasing availability of membrane performance data, allowing ML models to approximate highly non-linear phenomena that are difficult to describe using classical mechanistic formulations. However, this flexibility often comes at the cost of reduced interpretability, necessitating careful validation and, increasingly, the use of explainable-AI tools.

It should be emphasised that the application of ML in NF research remains constrained by practical limitations. Most published studies rely on relatively small, heterogeneous datasets and simplified performance metrics, which restrict model generalisability and the extent to which ML can yield robust mechanistic insight. Moreover, data-driven approaches in membrane science have not yet been fully supported by standardised practices for data curation, external validation, and reporting. Consequently, while ML has demonstrated strong predictive capability in NF applications, its role should presently be viewed as a complementary tool that enhances understanding and optimisation, rather than as a standalone substitute for physically grounded modelling frameworks (Figure 2) (Ignacz *et al.* 2025).

The diversity of ML models provides NF researchers with a flexible toolbox for addressing different data structures, prediction targets, and modelling objectives. Model selection in NF studies is therefore primarily governed by the nature of available data (e.g., static vs. time-series, laboratory vs. operational datasets), the required level of predictive accuracy, and the need for interpretability in relation to membrane transport and fouling mechanisms. Accordingly, ML approaches in NF are commonly categorised based on their learning paradigm, such as supervised, unsupervised, and reinforcement learning (RL). The following sections focus specifically on these model categories and their practical application in membrane science, with emphasis on NF membranes.

Despite the rapid expansion of ML applications in NF research, several constraints remain evident. Most published studies rely on limited and heterogeneous datasets, often derived from single laboratories or narrowly defined operating conditions, which restrict model robustness and generalisability. In addition, the transition towards data-driven modelling has not yet been matched by standardised practices for data curation, external validation, and transparent reporting in membrane research. Consequently, while ML has demonstrated strong predictive capability in NF applications, its ability to deliver broadly transferable mechanistic insight and plant-scale reliability remains only partially realised.

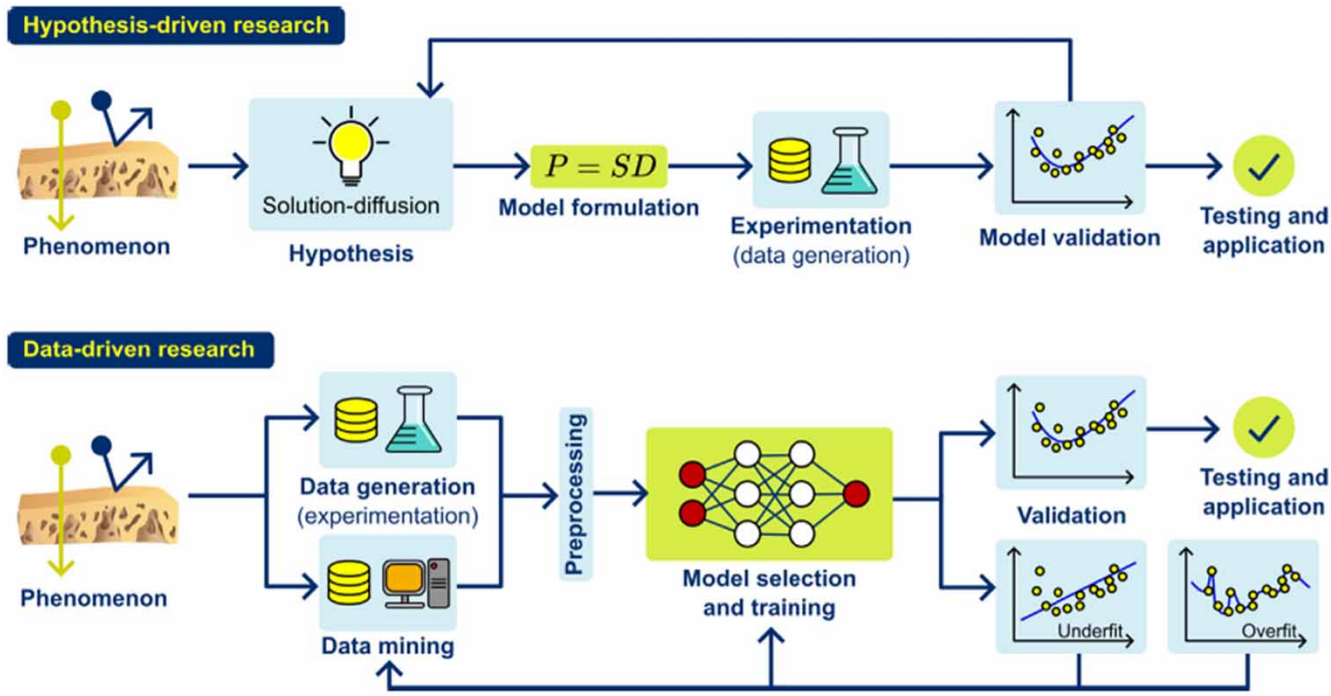


Figure 2 | Hypothesis-driven and data-driven research (Ignacz *et al.* 2025).

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

4.2. AI in membrane science

AI has become an important enabler of data-driven modelling in environmental and membrane engineering. Within membrane science, ML and DL are the most relevant AI branches, as they provide predictive and optimisation frameworks capable of capturing the highly non-linear behaviour of membrane processes (Cao *et al.* 2024; Shahouni *et al.* 2024). In NF systems, performance is governed by complex interactions among membrane properties, operating conditions, and solute characteristics, which often challenge classical modelling approaches. Conventional theories such as the solution–diffusion model (SDM) and Donnan exclusion principles offer valuable mechanistic insight but frequently fail to provide accurate predictions across diverse water matrices and operating regimes. AI-based models offer a complementary advantage by learning system behaviour directly from data, enabling the identification of non-linear patterns that are difficult to represent using simplified physical assumptions (Ignacz *et al.* 2023a). Whereas traditional NF studies have relied heavily on semi-empirical correlations derived under controlled laboratory conditions, ML models trained on heterogeneous datasets have demonstrated improved robustness when extrapolated to variable feed compositions and dynamic operating environments. For example, data-driven models have shown superior capability in predicting flux decline and fouling trajectories under fluctuating feed conditions, where mechanistic fouling models often lose validity (Hu *et al.* 2025; Singh *et al.* 2025). For example, whereas mechanistic fouling models often collapse under highly variable feed conditions, ML models trained on diverse datasets can successfully predict flux decline trajectories. In a broader sense, AI can be described as a form of *behavioural science of machines*. Just as behavioural science studies human decision-making under uncertainty, AI models exhibit learning behaviours shaped by feedback. In supervised learning, models behave by adjusting internal weights in response to labelled outcomes. In RL, the behaviour is refined by rewards and penalties, mirroring human trial-and-error learning (Cao *et al.* 2024). This behavioural analogy is particularly useful in NF research: membranes themselves respond dynamically to their environment (fouling, scaling, chemical changes), and AI provides a behavioural mirror – adapting predictions in real time as system conditions evolve.

Recent studies increasingly position AI within the broader digitalisation of water treatment systems. Shahouni *et al.* (2024) highlighted the role of AI in predictive modelling and membrane material discovery, while Cao *et al.* (2024) emphasised its integration into digital twin frameworks for membrane plants. In NF applications, AI-enabled digital twins provide dynamic representations of real systems, allowing prediction of flux, rejection, and fouling behaviour, as well as optimisation of cleaning and operational strategies. Practical applications demonstrate the maturity of these approaches: Zhang *et al.* (2021) applied deep representation learning to predict NF flux and rejection under limited data availability. Ji *et al.* (2025) employed artificial neural networks (ANNs) combined with explainable-AI tools to optimise NF performance for lithium recovery, and Gong *et al.* (2024) developed hybrid DL models to accurately predict perfluorooctane sulfonate (PFOS) rejection while maintaining interpretability. Collectively, these studies illustrate that AI in NF research serves not only as a predictive tool but also as a means to elucidate performance-relevant behavioural patterns under complex and variable conditions.

The reviewed studies (Table 2) collectively indicate that AI can enhance both predictive capability and process understanding in NF research; nevertheless, several NF-specific limitations remain. Most reported applications are confined to laboratory or pilot-scale systems, with model training commonly based on datasets derived from single studies or narrowly defined operating conditions. Such constraints limit model robustness and raise concerns regarding transferability to full-scale NF plants and to feedwaters with substantially different physicochemical characteristics.

Although explainable-AI techniques, such as SHAP, have improved the interpretation of data-driven models and provided insights into dominant NF performance drivers, their adoption remains inconsistent, and systematic benchmarking against established transport and fouling models is still limited. As a result, current AI applications in NF should be viewed as complementary tools that support performance prediction and optimisation, rather than as standalone replacements for physically grounded modelling frameworks.

4.3. Categories of ML

ML approaches applied in NF research are commonly categorised into supervised, unsupervised, and RL paradigms. This classification is not merely conceptual; recent NF studies demonstrate that the choice of learning paradigm strongly influences predictive accuracy, interpretability, and applicability across different tasks, including solute rejection prediction, fouling assessment, and resource recovery (Hu *et al.* 2021; Wang *et al.* 2024a; Singh *et al.* 2025).

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

Table 2 | Summary of studies applying ML and DL in NF applications (prepared by the researchers)

Algorithm	Technique	Use	Application in NF membranes	References
Random Forest (regression and classification)	Supervised learning with cross-validation, feature selection, and ROC analysis	Predicting permeability (Lp), selectivity ($1/S_0$), and trade-off coefficient (Tr); ranking membranes	PVDF, PES, PSf NF membranes fabricated by phase inversion; virtual screening via polySML	Liu *et al.* (2020)
BP-ANN + PSO	Deep ANN with PSO weight optimisation; 5-fold CV	Predicting NaCl rejection under varying operating and membrane conditions	NF desalination; NaCl removal efficiency prediction and optimisation	Zhang *et al.* (2024)
LSTM	Time-series DL (7-layer LSTM with dropout, ReLU)	Predicting flux decline and fouling layer thickness vs. time	NF fouling by humic acid, BSA, sodium alginate, tannic acid, and NOM mixtures	Shim *et al.* (2021)
1D-GBCNN	1D-CNN + gradient boosting; SHAP for feature importance	Predicting PFOS rejection: factor importance analysis	PFOS removal by different NF membranes (varying MWCO, flux, ion conc., pH)	Xiao *et al.* (2025)
XGBoost and CatBoost	Tree-based ensembles; SHAP; Bayesian hyperparameter optimisation	Predicting permeability, removal, flux decline/recovery, and reversible fouling	UF membranes (polymeric); methodology transferable to NF for fabrication and operation optimisation	Gao *et al.* (2023)
ANFIS, LR, SLR, MLR, SVM, and BT	Supervised ML and neuro-fuzzy; feature engineering	Predicting micropollutant rejection from 14 physicochemical, membrane, and operational variables	Self-prepared and commercial NF/RO membranes; NF-GP model identified key rejection drivers	Yogarathinam *et al.* (2024)
Genetic programming + deterministic optimisation	Evolutionary structure selection + regression; AIC-based model selection	Building predictive models for solute rejection: membrane rejection maps	Polymeric OSN membranes (PuraMem S600, 280); methodology applicable to NF solvent separations	Goebel *et al.* (2020)
XGBoost, LightGBM, AdaBoost, and CatBoost	Gradient boosting ensembles; SHAP; PDP	Predicting water permeance and salt rejection: key fabrication parameters	Interfacial polymerisation NF membranes; monomer concentration highlighted as dominant	Gong *et al.* (2024)
XGBoost and CatBoost + Bayesian optimisation	Tree-based ML with SHAP; Morgan fingerprints	Predicting permeability and salt rejection: inverse design of PA-based NF membranes	PA-based NF membranes; screened fabrication/monomer combinations beyond the current upper bound	Gao *et al.* (2022)
CNN, RNN, and RVM	1D-CNN, RNN, probabilistic regression; 10-fold CV	Predicting permeate conductivity, flow rate, and recovery	Hybrid NF/RO seawater desalination pilot plant; CNN best in all metrics	Abdu *et al.* (2025)
ANN, RNN, LSTM, CNN, FNN, RBFNN, and hybrid models	DL and hybrid ML (ANN + physics, GRU-FCN, etc.)	Predicting fouling (flux decline, TMP, resistance) and rejection	NF for drinking water and wastewater; DL outperformed mechanistic models; CNN + OCT for cake thickness	Zheng *et al.* (2024b)
GBR with molecular fingerprints	Gradient boosting; Frag, MACCS, Morgan, RDKitDP; SHAP and PDP	Predicting solvent permeance (P) and solute rejection (R): structure–performance relationships	OSN/TFC membranes for solvent recovery; framework extendable to NF membrane design	Wang *et al.* (2023c)
XGBoost with molecular fingerprints	Ensemble ML; feature engineering; SHAP; t-SNE	Predicting rejection of 228 organic solutes: key fragment identification	NF/RO membranes (polyamide, poly(piperazinamide), PES, CA); structure–rejection modelling	Lee *et al.* (2024)
RF, KNN, SVM, and BPNN	Supervised ML; PCA + correlation; SHAP; LOOCV	Predicting RDX crystal morphology and yield	NF-assisted cooling crystallisation (NF-CCr) with PI NF membranes (MWCO 200 Da)	Wang *et al.* (2025)

(*Continued.*)

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

Table 2 | Continued

Algorithm	Technique	Use	Application in NF membranes	References
QSAR, GA, VAE, GAN, inverse design	Data-driven and generative ML; digitalisation and miniaturisation	Roadmap for predictive and generative NF design: inverse design concepts	NF membranes in general; proposes PANPA, PAT integration, and open databases	Ignacz *et al.* (2023b)
XGBoost and CNN + Bayesian optimisation	DFT-assisted ML; quantum descriptors + process parameters; SHAP; t-SNE	Predicting permeance, dye and salt rejection: inverse design of selective membranes	Polyester NF membranes for textile wastewater recovery; high permeance–selectivity designs	Liu *et al.* (2025)
XGBoost with domain-knowledge embedding (DKD)	ML with mechanistic features (steric, charge, hydrophobicity); SHAP	Predicting and interpreting TrOC rejection: mechanism quantification	PA NF/RO membranes; NF more influenced by charge/hydrophobicity, RO by size exclusion	Wang *et al.* (2024b)
XGBoost (best), SVR, DTR, RF, AdaBoost, LightGBM	Supervised ML; Optuna Bayesian tuning; SHAP	Predicting protein rejection, steady flux, and permeance variation	UF membranes for protein purification; workflow transferable to NF process optimisation	Lu *et al.* (2025)
D-ANN (vs. S-ANN and 13 ML models)	Dual-output ANN; SHAP interpretation	Predicting relative permeability (RP) and relative rejection/selectivity (RR)	Thin-film nanocomposite NF membranes (TFNNMs) for OSN; key NP and fabrication factors identified	Li *et al.* (2023)
LSTM and LSTM–CSA	LSTM + Crow Search; GA/PSO-based feature selection; Monte Carlo	Predicting permeate conductivity: quantifying uncertainty	Hybrid NF–RO desalination; improved accuracy and robustness vs. standalone LSTM	Abba *et al.* (2023)
Residual three-component NN (R-TNN)	Deep NN with residuals; selective regularisation; PDP	Predicting relative water permeability (RWP) and relative salt passage (RSP)	Thin-film nanocomposite RO/NF membranes; design rules for breaking permeability–selectivity trade-off	Li *et al.* (2024)
NN, GPR, RF, XGBoost, CatBoost, CNN, DRL	ML + XAI (SHAP, PDP); inverse design; fingerprint + image features	Flux/rejection prediction: feature–performance analysis; ML-guided screening	NF highlighted: salt rejection & flux of polyamide NF; CNN/DRL for nanoporous graphene NF; BO-guided NF design	Cao *et al.* (2024)
Multimodal DL (ANN, 1D-CNN, 2D-CNN)	Fusion of adsorption + dynamic membrane data; Bayesian optimisation; SHAP	Predicting normalised water flux; analysing adsorption–filtration coupling	Dynamic membranes (polymer/ceramic + adsorbents); design insights for fouling mitigation	Jeong *et al.* (2024)
GRU	Recurrent DL; PCA for DOM features; pre-processing (outlier removal, smoothing)	Predicting TMP dynamics and fouling behaviour	Pilot NF systems (conventional-NF vs. O_3-BAC-NF); accurate TMP trend prediction for cleaning optimisation	Niu *et al.* (2025)
ANN, CNN, RNN, SVM, GA, PSO, ANFIS, hybrids	AI toolbox (NNs, DL, optimisation, neuro-fuzzy); PCA; CFD coupling	Predicting flux, fouling, and micropollutant dynamics: operating condition optimisation	NF, RO, FO, UF; review of AI tools with emphasis on NF fouling prediction and flux modelling	Shahouni *et al.* (2024)
RF, XGBoost, DMN, Bayesian optimisation	Multiscale (CB, MD, CG) + ML; hybrid atomistic–continuum; SHAP	Linking nanoscale interactions to flux/rejection: NF selectivity design	NF/RO membranes; ML-enhanced multiscale frameworks for NF design	Zhu *et al.* (2024)
LSTM, GA, Siamese CNN	LSTM for time-series, GA for optimisation, Siamese CNN for vision	Predicting water quality; fouling detection; cleaning schedule optimisation	Hybrid NF system (sedimentation + hybrid membranes + GO); >99% removal of metals and microbes	Parmar & Dhruv (2021)

(*Continued.*)

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

Table 2 | Continued

Algorithm	Technique	Use	Application in NF membranes	References
ANN, GBR, LightGBM, XGBoost, CatBoost, RF, SVM	AutoML (PyCaret); RDKit fingerprints + descriptors; SHAP	Predicting water flux (Jw) and rejection (R%) of micropollutants	FO membranes; workflow and insights transferable to NF/RO performance optimisation	Jafari *et al.* (2025)
RF, XGBoost	Ensemble ML; Bayesian optimisation; SHAP	Predicting Li^+ permeation and Li/Mg selectivity	NF membranes with mixed charge (PLIP-MOF, CAIP-MOF); MWCO and WCA identified as key features	Zheng *et al.* (2024a)
XGBoost, NGBoost–XGBoost	Gradient boosting; SHAP; probabilistic predictions	Predicting micropollutant removal; analysing size, electrostatics, adsorption	NF/RO membranes; highlighted data leakage issues and limited generalisability to unseen systems	Jeong *et al.* (2021)
RF, MLP, MLR	Supervised ML; hyperparameter tuning; SHAP	Predicting PWP, Na_2SO_4, and NaCl rejection; key synthesis parameters	TFC polyamide NF membranes; PIP and SLS concentrations are most influential	Sutariya *et al.* (2025)
SVR, ANFIS, RLR, MLR	Supervised ML; feature selection; grid search; membership functions	Predicting flux and rejection, including dye/salt fractionation	Loose NF membranes (Ag-PDA polysulfone); accurate flux/rejection and dye/salt separation predictions	Baig *et al.* (2023b)
RF, XGBoost	Ensembles; feature importance (MDI, loss); PDP	Predicting water/salt selectivity; ranking structural vs. operational parameters	Polyamide NF membranes; pore radius and zeta potential dominate salt rejection	Ma *et al.* (2023)
RF, XGBoost	Ensembles; permutation importance; univariate/bivariate PDP	Decoupling structural and operational effects on mono/divalent ion selectivity	Polyamide NF; $R^2 > 0.78$–0.99 for various ion pairs; mechanistic insights into sieving vs. Donnan exclusion	Lu *et al.* (2023)
Linear regression, RF, BT, XGBoost	Supervised ML; cross-validation; SHAP	Predicting permeability and selectivity of LBL NF membranes; screening polymers	LBL NF membranes; 23 promising polymers shortlisted via ML screening	Wang *et al.* (2024a)
BT, SVM, ANN, LR	Supervised ML; PDP	Predicting relative permeability and selectivity of TFN-OSN membranes	TFN membranes for OSN; identified optimal NP loading and fabrication parameters	Wang *et al.* (2023a)
MLR, SVM, ANN, kNN, RF, GBDT, XGBoost, LightGBM	Supervised ML; SHAP-based feature selection; grid search CV	Predicting rejection of 276 organic contaminants: feature ranking	NF/RO membranes; XGBoost-10 best; steric effects (MWCO, V, MW) dominant, with adsorption and electrostatics secondary	Zhu *et al.* (2023)
LR, Ridge, LASSO, SVR, MLP, KNN, DT, AdaBoost, ET, Bagging, RF	Comparative supervised ML; 5-fold CV; SHAP	Predicting permeability, MWCO, zeta potential, Li/Mg selectivity	TFC NF membranes for Li/Mg separation; $MgCl_2$ rejection most influential for selectivity	Sun *et al.* (2025)
Neuro-Fuzzy, LSSVM, SAE, NF-based non-linear ensemble	Hybrid ensemble ML; sensitivity analysis; k-fold CV	Predicting dye/divalent salt rejection (RJDS %)	Loose NF membranes (Ag-PDA supports); NF-M2 and SAE-NL ensembles gave near-perfect accuracy	Baig *et al.* (2023a)
GNN + PLS	GNN with solute/solvent structures; Bayesian optimisation; explainable AI	Predicting solute rejection: solvent effect analysis	Polyimide OSN (DuraMem 300); methodology transferable to aqueous NF	Ignacz *et al.* (2023a)

(*Continued.*)

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

Table 2 | Continued

Algorithm	Technique	Use	Application in NF membranes	References
MLR, SVM, XGBoost	Supervised ML; Optuna optimisation; SHAP, ICE, PDP	Predicting OMP rejection: identifying dominant mechanisms	NF/RO polyamide membranes; vdW radius, size ratio, and charge product critical; MWCO 200–300 Da recommended	Hu *et al.* (2025)
RF, XGBoost	Ensemble ML; SHAP & PDP	Predicting dye rejection and salt permeability: structural vs. operational effects	Loose NF membranes; structural parameters (pore radius, zeta potential) are more critical than operational parameters	Zhang *et al.* (2025)
ANN, RF, XGBoost	ANN (4-layer MLP), RF, XGBoost with grid search	Predicting micropollutant removal efficiency	NF/RO membranes; MW and Kow dominant; pressure improved removal efficiency	Masuodi *et al.* (2024)
ANN, SVM, RF	Supervised ML with PCA; large multi-study dataset	Predicting permeance and rejection; dominant descriptor identification	Commercial OSN membranes; framework applicable to NF/RO data integration	Hu *et al.* (2021)
RF, GBM, AdaBoost (plus SVR, MLP, LR, RR)	Ensemble ML; 5-fold CV; grid search; permutation importance	Predicting PFOS rejection: identifying key operational factors	NF membranes for PFOS removal; pressure, initial PFOS concentration, membrane type, and pH highlighted	Hosseinzadeh *et al.* (2022)
MLR, SVM, MLP, RF, XGBoost	Supervised ML with grid search CV; SHAP	Predicting rejection of regulated and unregulated HAAs	NF membrane (VNF2); solute size (rs, logKow, MW) and pH/TMP identified as key factors	Wang *et al.* (2023b)
MLR, Lasso, Ridge, RF, ANN	Supervised ML; normalisation, one-hot encoding, outlier removal; ANOVA	Predicting PFOS rejection under multifactorial conditions	NF membranes (NF90, NF200, NF270, NF245, NF400, NF500); ANN and RF identified optimal pH 6–8 and 25 °C	Singh *et al.* (2025)
XGBoost, CatBoost, RF, GB, SVR, MLP + PSO	Ensemble ML + PSO; MICE imputation; SHAP, PDP, ICE; Sobol analysis	Predicting Li^+ permeation and Li^+/Mg^{2+} selectivity; multi-objective optimisation	NF membranes for lithium extraction; optimum MWCO 350–400 Da and WCA 20–40 °C	Ji *et al.* (2025)
DNN, RF, SVR, GBR	DNN with data augmentation; representation learning	Predicting salt rejection and water flux with small datasets	Polyamide TFC NF membranes; augmented data markedly improved prediction accuracy	Zhang *et al.* (2021)
ANN with Bayesian regularisation	ANN with seven inputs; neuron optimisation; sensitivity analysis	Predicting solute rejection, PWF, and flux recovery	Nanocomposite membranes (UF focused); universal design platform extendable to NF/RO	Fetanat *et al.* (2021)

4.3.1. Supervised learning

Supervised learning is the most widely adopted ML paradigm in NF research, as it directly links measurable inputs to performance outputs using labelled datasets. In NF applications, inputs typically include operating conditions (e.g., transmembrane pressure, pH, ionic composition, solute MW) and membrane properties (e.g., pore size, surface charge, hydrophilicity), while outputs represent performance indicators such as water flux, solute rejection, fouling indices, or energy consumption. Numerous studies have demonstrated the effectiveness of supervised learning for predicting NF performance. For example, Wang *et al.* (2023b) applied extreme gradient boosting to predict haloacetic acid rejection, achieving high predictive accuracy while identifying dominant rejection drivers. Singh *et al.* (2025) showed that ANNs outperformed other models in predicting PFOS rejection, whereas random forest (RF) models provided valuable interpretability through variable-importance analysis. Similarly, Hu *et al.* (2025) employed supervised models to predict pharmaceutical compound rejection and to elucidate the relative roles of size exclusion and electrostatic interactions.

Despite these successes, supervised learning in NF remains constrained by data availability and quality. Model performance is strongly dependent on the representativeness of labelled datasets, which are often limited by experimental cost and laboratory-specific protocols. Moreover, validation is frequently based on random data splits from single datasets,

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

offering limited evidence of external validity. Without systematic testing across independent datasets or different NF systems, the generalisability of supervised models remains uncertain.

4.3.2. Unsupervised learning

Unsupervised learning aims to identify hidden structures and patterns in data without relying on labelled outputs. This paradigm is particularly relevant for NF research when dealing with high-dimensional water quality data, membrane characterisation datasets, or long-term operational records. Although less frequently applied than supervised learning, recent studies highlight its potential value in NF-related investigations. Ignacz *et al.* (2023a) demonstrated that clustering and dimensionality reduction techniques can reveal meaningful groupings of membrane–solute interactions in OSN, a concept transferable to aqueous NF systems. Niu *et al.* (2025) applied principal component analysis to characterise dissolved organic matter (DOM) affecting NF fouling, identifying key components that explained the majority of data variance and simplifying subsequent modelling efforts.

The primary strength of unsupervised learning in NF lies in its exploratory capability, enabling identification of dominant patterns, operational regimes, or membrane performance clusters. However, because it does not provide direct performance predictions, unsupervised learning is most effective when used in combination with supervised approaches. At present, many unsupervised NF studies remain exploratory, with limited integration into predictive or control frameworks. In addition, results are often sensitive to data pre-processing choices and clustering metrics, underscoring the need for more rigorous evaluation of stability and physical interpretability before such insights are used to guide membrane design or operation.

Regardless of the learning paradigm employed, successful ML deployment in NF requires careful model selection, validation, and interpretation, as summarised schematically in Figure 3.

4.3.3. Reinforcement learning

RL represents a distinct paradigm in ML, where an agent interacts with an environment and improves its decision-making through rewards and penalties. Unlike supervised learning, which requires labelled input–output pairs, RL is well-suited for sequential optimisation tasks and real-time process control. In the field of membrane technologies, RL has only recently been introduced, yet several pioneering studies demonstrate its considerable potential. Bonny *et al.* (2022) presented one of the earliest applications of deep RL for RO desalination. Their study developed a deep RL-based control system capable of dynamically adjusting operating parameters such as feed flow and pressure to maximise water recovery while minimising energy consumption. The system demonstrated superior adaptability compared with fixed-parameter control strategies, effectively handling variations in feedwater salinity and demonstrating the feasibility of RL in real-time process optimisation.

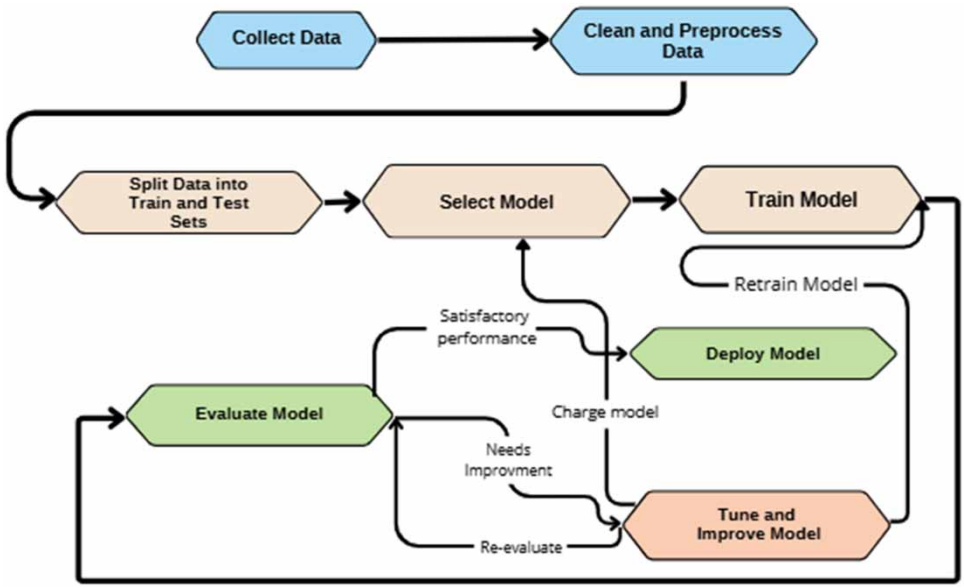


Figure 3 | Key steps of supervised/unsupervised model selection, training, and deploying (prepared by the researchers).

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

Yoon *et al.* (2022) applied RL to automate membrane capacitive deionisation (MCDI), optimising regeneration cycles and energy efficiency. By training the RL agent to balance adsorption and desorption phases, the model achieved a reduction in energy demand without compromising salt removal efficiency. Similarly, Park *et al.* (2022) explored the use of deep RL in UF systems, focusing on optimising transmembrane pressure and cleaning intervals. Their results revealed that RL could autonomously identify cleaning schedules that maintained stable flux for longer periods while reducing chemical cleaning frequency, thereby extending membrane lifespan and lowering operational costs.

More recently, Golabi *et al.* (2024) implemented a deep RL framework for optimal RO operation. Their system balanced permeate quality and energy use under fluctuating operating conditions, outperforming conventional control models. By learning adaptive strategies, the RL agent could maintain performance targets even when confronted with disturbances such as feedwater quality changes, highlighting RL's potential for integration into advanced digital twin environments.

Collectively, these studies confirm that while RL remains in its infancy within NF and RO research, it offers clear advantages for autonomous, adaptive, and multi-objective optimisation. The capacity of RL to learn from feedback and continuously refine operational strategies mirrors the dynamic challenges of membrane processes, where fouling, scaling, and fluctuating water quality demand flexible and intelligent responses. However, RL applications remain largely at the proof-of-concept stage, with real-world deployment requiring further validation, especially regarding stability, safety, and data requirements in complex NF/RO systems (Cao *et al.* 2024). The examples summarised here show that RL offers a promising paradigm for autonomous process optimisation; nevertheless, current work remains at the proof-of-concept stage. Most RL studies rely on simulated environments or tightly controlled pilot plants, and questions related to stability, safety constraints and fault tolerance in full-scale NF/RO systems have not yet been comprehensively addressed. Furthermore, the data and computational requirements for training RL agents are substantial, which may limit adoption in typical water utilities. Future research will therefore need to focus on the design of safe, sample-efficient RL schemes and their integration into regulatory-compliant digital twin frameworks.

4.3.4. Comparative perspective

Taken together, the three categories of ML – supervised, unsupervised, and RL serve complementary and increasingly synergistic roles in NF research. Supervised learning remains the dominant paradigm, excelling in predictive modelling when sufficient labelled data is available, as demonstrated in studies of PFOS rejection, pharmaceutical removal, and HAA behaviour. Unsupervised learning, while less widely applied, provides an important exploratory function by uncovering hidden patterns in complex datasets, revealing clusters of solutes with similar rejection profiles or operational regimes associated with fouling risk. RL, although still largely conceptual, offers an adaptive layer capable of enabling real-time optimisation within digital twin frameworks by learning from continuous feedback. Recent reviews stress that the true power of ML in NF does not lie in these paradigms in isolation, but rather in their integration. A realistic future scenario is one in which unsupervised learning is used to classify and reduce complexity in large datasets, supervised models then predict key outputs such as rejection or fouling under identified regimes, and RL adaptively fine-tunes operational strategies in real time. Such a layered approach mirrors human learning behaviours, combining classification, prediction, and adaptation, and resonates strongly with the adaptive response of NF membranes themselves to dynamic environments. The implication is that NF research is moving towards not just isolated predictive models, but towards comprehensive, self-learning frameworks that integrate multiple ML paradigms into a single intelligent system.

4.4. ML algorithms and interpretability studies

ML algorithms are the backbone of data-driven research in NF. Their success depends not only on predictive accuracy but also on robustness, interpretability, and applicability across diverse datasets. In recent years, NF literature has seen a wide spectrum of ML algorithms, ranging from regression-based models to advanced ensemble methods and neural networks. Each of these has demonstrated strengths and limitations depending on dataset size, quality, and complexity.

4.4.1. Regression-based models

Regression-based methods represent the earliest ML applications in NF research. They establish mathematical relationships between input variables (e.g., pressure, solute concentration, pH) and output performance metrics (e.g., flux, rejection). Their simplicity provides interpretability, but their inability to model non-linear interactions often limits predictive accuracy.

Singh *et al.* (2025) compared linear regression and ridge regression with advanced models such as RF and ANN. Using datasets of PFOS rejection under varying pressures (2–8 bar), pH values, and ionic compositions, they observed that

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

linear regression achieved only moderate predictive power ($R^2 \approx 0.78$). Ridge regression slightly improved the fit by penalising large coefficients but still lagged non-linear models. The authors concluded that regression methods could identify general trends, such as the positive effect of higher pressures on rejection, but failed to capture interaction effects such as the combined influence of divalent cations and pH on PFOS rejection.

Ignacz *et al.* (2023a) also tested regression models in their work on OSN and highlighted similar shortcomings. While regression models were interpretable and aligned with solution–diffusion theory, they consistently underestimated flux under conditions of high solute loading, demonstrating that linear assumptions break down in highly non-linear systems. Regression models remain useful for establishing baselines and for situations where interpretability is prioritised. Their ability to clearly show linear relationships makes them valuable for educational or initial exploration stages. However, the NF literature consistently demonstrates that regression methods underperform compared with non-linear ML methods, particularly for datasets with interacting operational and material variables. Overall, regression-based models retain value as transparent baselines and for generating first-order insights into variable relationships; however, their limitations are consistently exposed in NF applications involving strong non-linearity and interaction effects. The tendency of such models to underestimate performance at the extremes of operating ranges, and their inability to represent complex coupled phenomena such as simultaneous fouling and speciation changes, suggest that their role in modern NF research should be primarily supportive. In many cases, regression models are most useful when embedded within hybrid or physics-informed frameworks rather than used as standalone predictive tools.

4.4.2. Tree-based ensemble methods (RF, GBM, XGBoost)

Tree-based ensemble methods are among the most successful ML approaches applied in NF. Algorithms such as RF, gradient boosting machine (GBM), and extreme gradient boosting (XGBoost) combine multiple decision trees to improve predictive accuracy and reduce overfitting. Importantly, they also provide variable importance rankings, which offer mechanistic insights into NF processes. Figure 4 illustrates an ensembled learning approach.

Singh *et al.* (2025) applied RF alongside ANN, SVM, and regression to predict PFOS rejection. RF models achieved $R^2 \approx 0.91$, slightly lower than ANN but still strong. Importantly, RF identified pH and trivalent cations as the most influential parameters, aligning with experimental knowledge that PFOS rejection is highly sensitive to ionic composition. The authors

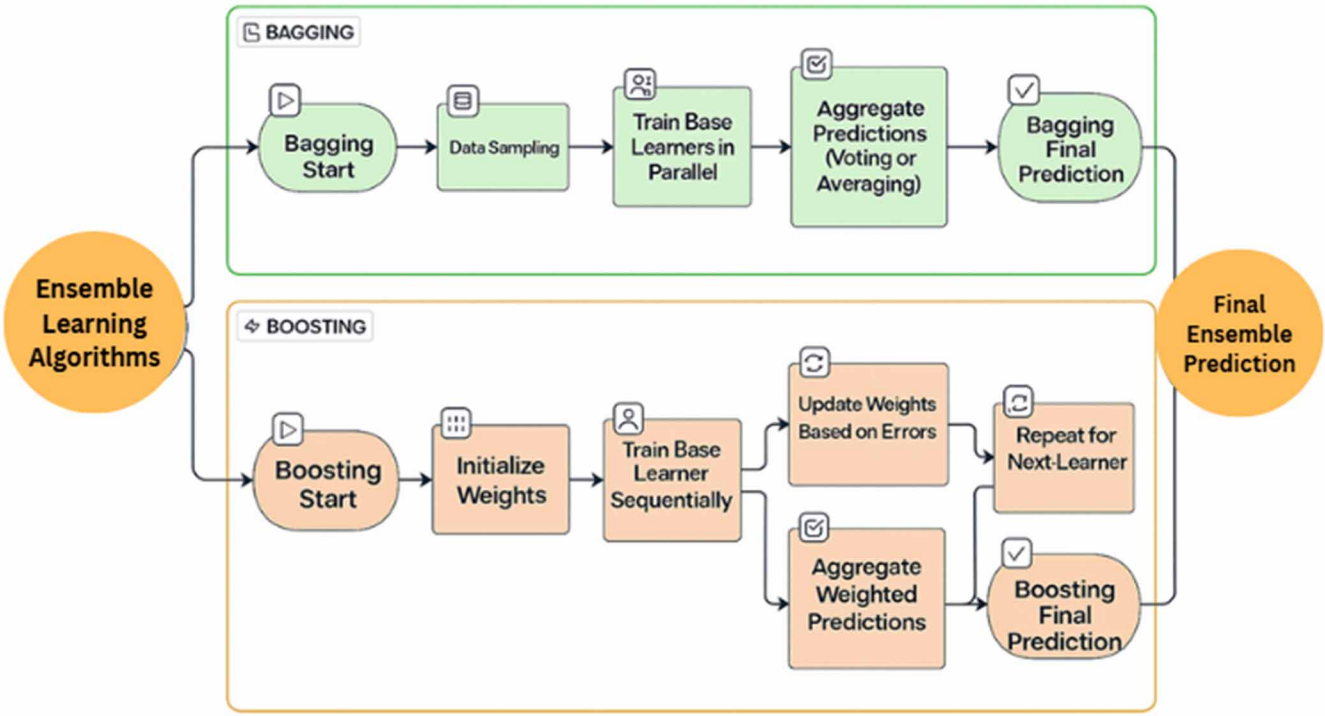


Figure 4 | Ensembled learning approach.

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

concluded that while ANN offered higher raw accuracy, RF provided clearer interpretability, making it valuable for mechanistic understanding. Similarly, Hu *et al.* (2025) used RF, ANN, and XGBoost to model the rejection of pharmaceutical active compounds (PhACs), including antibiotics and endocrine-disrupting chemicals. XGBoost slightly outperformed the other methods ($R^2 > 0.9$ overall), and feature importance analysis revealed that solute molecular size and charge were the dominant rejection factors, while hydrophobicity was less influential for neutral compounds. This result challenged earlier assumptions that hydrophobicity was always a primary determinant of rejection.

These examples highlight that ensemble methods consistently balance predictive accuracy with interpretability. While ANN may sometimes surpass RF or XGBoost in predictive accuracy, the variable importance outputs of ensemble models provide mechanistic insights crucial for scientific acceptance. For instance, ensembles frequently identify solute–membrane charge interactions as more influential than operating pressure, a pattern repeatedly validated across NF studies. Thus, ensembles function not only as predictive tools but also as knowledge discovery frameworks, guiding researchers towards the most critical variables influencing NF performance. Despite their strong performance, ensemble methods are not without drawbacks. Their apparent robustness can conceal overfitting when datasets are small or when data leakage is not carefully avoided, and the physical interpretability of variable importance rankings can be overstated if correlated inputs are not properly treated. In addition, many studies compare ensembles and ANNs on a single dataset without rigorous statistical testing or external validation, making it difficult to generalise claims about superiority. These issues highlight the need for more systematic benchmarking of ensemble models across multiple NF case studies and for clear reporting standards regarding feature selection and model validation.

4.4.3. Artificial neural networks

ANNs are among the most widely applied ML tools in NF research because of their ability to approximate highly non-linear relationships between operational parameters, solute characteristics, and membrane performance.

Ji *et al.* (2025) applied ANN models to predict lithium permeation and Li/Mg selectivity in NF membranes treating unconventional water sources. Their ANN achieved $R^2 > 0.96$, outperforming regression and RF models. Using Shapley Additive Explanations (SHAP), they demonstrated that membrane MW cut-off and contact angle had greater influence on selectivity than operating parameters such as pressure. This result not only validated the ANN's predictions against physical intuition but also showcased how explainability tools can mitigate the 'black-box' perception of neural models.

Zhang *et al.* (2021) employed deep representation learning, with ANN-based components, to model the performance of polyamide thin-film NF membranes. To address data scarcity, they used data augmentation strategies to artificially expand the dataset. Their ANN achieved mean relative errors below 5% for both flux and rejection, proving that even under limited data availability, ANN performance can be sustained when combined with augmentation techniques.

Hu *et al.* (2025) tested ANN alongside RF and XGBoost for predicting PhAC rejection. ANN achieved R^2 values in the 0.92–0.95 range, comparable to XGBoost and superior to SVM. The model captured the importance of electrostatic interactions in rejection mechanisms, aligning with observed experimental results. However, the authors cautioned that ANN models are prone to overfitting with limited data, highlighting the need for careful regularisation.

Collectively, ANNs consistently deliver high predictive accuracy in NF studies, outperforming regression and often matching ensemble models. Their main weakness is overfitting and limited interpretability, but explainability frameworks such as SHAP and sensitivity analysis are increasingly adopted to improve trust in their predictions. While ANNs consistently deliver high predictive accuracy, their widespread use in NF research raises several methodological concerns. Overfitting remains a recurrent issue, particularly when complex architectures are trained on limited datasets that contain experimental noise or unbalanced operating conditions. Regularisation strategies and data augmentation are not always reported in sufficient detail, and uncertainty quantification is seldom undertaken. Furthermore, although explainable-AI tools have begun to reduce the black-box character of ANNs, the correspondence between attribution scores and true mechanistic causality is not always critically examined. Consequently, ANN-based findings should be interpreted cautiously unless supported by independent experimental validation or mechanistic reasoning.

4.4.4. Support vector machines (SVMs)

SVMs are effective for high-dimensional and small-sample problems, but their use in NF has been limited compared with ANN and ensemble methods. Singh *et al.* (2025) included SVM in their comparative analysis of PFOS rejection. The SVM achieved $R^2 \approx 0.82$, which was lower than RF (0.91) and ANN (0.95). The authors suggested that SVM struggled

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

with the non-linear interactions between pH, ionic strength, and pressure, which are common in NF. Hu *et al.* (2025) also tested SVM for pharmaceutical rejection. Their results indicated that SVM achieved moderate performance ($R^2 \approx 0.85$), but lagged behind XGBoost and ANN. They concluded that SVM could be valuable when datasets are small and simple, but it is less suitable for large, complex datasets with multiple interacting parameters. SVM has value as a benchmark model, especially in early-stage research or with limited datasets. However, NF literature shows a clear trend: as datasets grow in size and complexity, ensemble methods and ANN consistently outperform SVM in both accuracy and robustness. The comparative studies reviewed suggest that SVMs now occupy a secondary role in NF applications. Their advantages for small and moderately complex datasets are increasingly overshadowed by the flexibility of ensemble and neural network methods. Moreover, the sensitivity of SVM performance to kernel choice and hyperparameter tuning is rarely explored systematically, and the resulting models often provide limited mechanistic insight. In practice, SVMs function mainly as useful benchmarks rather than as methods of choice for state-of-the-art NF modelling, and future work may justifiably prioritise more interpretable or more scalable alternatives.

4.4.5. Hybrid and ensemble frameworks

Hybrid and ensemble frameworks combine the strengths of multiple algorithms, improving accuracy, robustness, and sometimes interpretability. Ji *et al.* (2025) integrated ANN with multi-objective optimisation to balance lithium permeation and Li/Mg selectivity. Their hybrid framework revealed trade-offs between high lithium recovery and maintaining selectivity, providing practical design insights for NF-based resource recovery. Wang *et al.* (2023b) demonstrated that XGBoost, an ensemble boosting algorithm, outperformed ANN and SVM in predicting HAA rejection ($R^2 > 0.95$). Similarly, Hu *et al.* (2025) confirmed that ensemble models consistently provided higher robustness across varying water chemistries. Hybrid and ensemble methods are increasingly viewed as state-of-the-art in NF-ML applications. They combine the predictive accuracy of deep models with the interpretability of ensemble trees, offering both performance and mechanistic insights.

Hybrid and ensemble frameworks clearly represent a methodological frontier for NF research, yet their complexity also introduces challenges. The combination of multiple models can obscure error sources and make it difficult to diagnose why a given framework succeeds or fails under specific conditions. Moreover, the computational and implementation costs of such frameworks may limit their adoption in routine engineering practice, particularly in resource-constrained settings. To ensure that these advanced approaches translate into real-world impact, future studies should report not only accuracy metrics but also computational requirements, implementation complexity, and comparative performance against simpler baselines.

Taken together, the comparative analysis presented in Table 3 indicates that no single ML approach is universally optimal for NF applications. Tree-based ensemble models and shallow neural networks currently dominate NF research due to their strong performance with small-to-medium tabular datasets, high predictive accuracy, and, in the case of ensemble methods, favourable interpretability through feature-importance analysis. DL architectures, including recurrent and convolutional neural networks (CNNs), exhibit clear advantages in handling time-dependent fouling dynamics and imaging-based analyses, respectively; however, their application remains constrained by data availability, computational demand, and limited interpretability.

Across all algorithm families, persistent methodological challenges are evident, including reliance on limited and heterogeneous datasets, inconsistent validation strategies, insufficient treatment of data leakage, and scarce uncertainty quantification. Addressing these issues will require a shift towards benchmark-oriented evaluation practices, supported by open and standardised datasets, transparent validation protocols, and systematic uncertainty analysis. Such developments are essential for translating current algorithmic advances into robust and generalisable tools for NF membrane design, process optimisation, and plant-scale operation.

The observed performance differences among ML models in NF applications can be largely attributed to the nature of NF processes and the structure of available datasets. Tree-based ensemble models frequently outperform linear approaches because NF performance metrics such as flux, solute rejection, and ion selectivity are governed by non-linear interactions among membrane structural parameters (e.g., pore size, surface charge, hydrophilicity), solute properties, and operating conditions. Ensemble methods effectively capture these interactions while remaining robust to multicollinearity and noise, which are common features of experimental NF datasets.

ANNs demonstrate superior performance when sufficient data are available, as their layered architectures enable flexible approximation of complex multi-variable relationships. In NF studies, ANNs are particularly effective for predicting

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

Table 3 | Comparative performance of ML algorithms for NF applications (prepared by the researcher)

Algorithm family	Best performing conditions	Typical NF applications	Strengths	Limitations/challenges
Linear and regression models (MLR, Ridge, LASSO)	Small datasets; predominantly linear relationships	Baseline flux/rejection prediction; sensitivity analysis	High interpretability; low computational cost	Poor performance on non-linear interactions; limited accuracy for complex NF systems
Tree-based ensembles (RF, GBM, XGBoost, LightGBM)	Small–medium tabular datasets; heterogeneous physicochemical variables	Flux/rejection prediction; ion selectivity; parameter importance analysis	Strong accuracy; captures non-linearity; SHAP interpretability	Can overfit noisy datasets; dependent on good feature engineering
Shallow neural networks (ANN/MLP)	Medium–large datasets; non-linear multi-variable interactions	Predicting rejection, permeability, fouling indices	High predictive accuracy; adaptable to complex datasets	Limited interpretability; prone to overfitting with small data
Recurrent neural networks (LSTM, GRU)	Sequential/time-series data; fouling development over time	Flux decline modelling; TMP evolution; fouling thickness prediction	Excellent for temporal dynamics; captures long-term dependencies	Needs larger datasets; high computational cost
Convolutional neural networks (CNNs)	Imaging, structural, or spatial datasets (e.g., OCT images)	Fouling layer detection; microstructure–performance mapping	Strong at feature extraction; high-resolution predictions	Requires specialised data; may lack interpretability
Hybrid models (CNN–Boosting, ANN + Optimisation, Physics-informed ML)	Multimodal or complex datasets; need for accuracy + interpretability	PFOS removal; membrane design optimisation; inverse design	Combine accuracy, robustness, and insight; versatile	More difficult to implement; higher data and computational requirements
Reinforcement learning (Deep RL)	Dynamic operational control; real-time optimisation	Pressure optimisation; cleaning scheduling; NF/RO plant control	Learns adaptive strategies; suitable for digital twins	Still experimental; limited real plant validation

steady-state flux and rejection behaviour where interactions are strong but temporal dependencies are limited. However, their reduced interpretability and sensitivity to dataset size constrain their applicability in data-scarce or exploratory studies.

DL architectures outperform shallow models primarily in applications involving temporal or spatial structure. Recurrent neural networks (RNNs), such as long short-term memory (LSTM) and gated recurrent unit (GRU), excel in modelling, fouling dynamics and flux decline because they explicitly account for time-dependent dependencies and system memory effects inherent to fouling and cleaning cycles. CNNs achieve superior performance in image-based NF applications by extracting spatial features from membrane morphology or fouling layer visualisation data. These advantages explain the increasing adoption of DL in digital twin frameworks and dynamic NF system monitoring, despite their higher data and computational requirements.

5. DL: ARCHITECTURES, APPLICATIONS, AND CHALLENGES

5.1. Introduction to DL

DL is a specialised subset of ML that employs multi-layered neural networks capable of extracting hierarchical and abstract representations from data. Unlike traditional ML algorithms, which rely heavily on engineered features, DL methods can automatically learn feature representations directly from raw or minimally processed inputs. This makes them particularly effective for handling large, complex, and non-linear datasets. For NF, DL has become increasingly relevant in modelling flux behaviour, solute rejection, fouling dynamics, and membrane material properties. NF systems are highly non-linear due to the interplay of pressure, solute chemistry, water chemistry, and membrane structure. DL's capacity to learn from such complex interactions makes it an ideal candidate for advancing predictive modelling in this field.

5.2. Convolutional neural networks

CNNs are designed to extract spatial features, making them effective for image-based or structurally complex data. In NF, CNNs have been employed for both material modelling and fouling analysis. Abdu *et al.* (2025) designed a real-time

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

hybrid NF/RO performance prediction framework for seawater desalination plants using DL. Their model, which incorporated CNN-based feature extraction alongside feed-forward layers, predicted permeate flux, salt rejection, and energy consumption with R^2 values above 0.95. Beyond accuracy, the framework demonstrated adaptability to variations in feedwater salinity and transmembrane pressure, making it suitable for real-time plant optimisation. The authors highlighted the potential of such CNN-enhanced systems to serve as the computational backbone of digital twins, providing operators with predictive insights and control recommendations that reduce fouling risk and energy demand.

Complementing this, Shahouni *et al.* (2024) reviewed recent advances in applying AI, including CNNs, to predict membrane behaviour in water purification and desalination. They reported that CNNs are particularly effective in capturing non-linear solute–membrane interaction patterns when datasets include diverse operational conditions and solute properties. Compared with SVMs and regression methods, CNN-based frameworks consistently delivered higher accuracy and better generalisation, while emerging explainability techniques were noted as essential for improving trust in their predictions. These studies illustrate how CNN-driven DL is shifting from experimental modelling towards plant-scale deployment. By enabling accurate, real-time, and interpretable predictions, CNN frameworks represent a crucial step towards fully intelligent NF/RO desalination systems.

5.3. RNNs, LSTM, and GRU

RNNs and their variants, LSTM and GRU, are designed to capture temporal dependencies in sequential data. In NF, they are especially useful for modelling flux decline and fouling development over time. Niu *et al.* (2025) applied LSTM and GRU models to predict fouling caused by DOM. Their study demonstrated that recurrent architectures achieved $R^2 > 0.93$, outperforming ANN and RF models. LSTM's 'memory cells' allowed it to capture long-term fouling trends that regression and RF models missed. The GRU model, while slightly less accurate, offered computational efficiency, making it suitable for real-time applications. The authors concluded that RNNs provide an effective tool for predicting time-dependent NF performance, especially for fouling control. RNN-based models add unique value in NF research because fouling processes are inherently dynamic. Their ability to retain memory of prior states aligns with the physical reality of NF, where fouling accumulates progressively.

5.4. Multimodal and hybrid DL approaches

As NF applications expand, researchers increasingly recognise that single-source datasets are insufficient to capture the complex interplay of factors governing membrane performance. Multimodal DL, the integration of heterogeneous data sources such as operational parameters, solute descriptors, and membrane structural properties has emerged as a promising direction. By combining complementary modalities, models can learn richer representations that mirror the multidimensional nature of NF processes. Shahouni *et al.* (2024) highlighted the promise of multimodal approaches in their review of AI for water purification. They argued that integrating physicochemical descriptors of solutes with operational data enables models to capture both intrinsic rejection mechanisms (e.g., size exclusion, electrostatic interactions) and external influences (e.g., pH, ionic strength).

Beyond multimodal data fusion, hybrid DL frameworks combine different neural architectures or integrate DL with ensemble methods to achieve both accuracy and interpretability. Xiao *et al.* (2025) exemplified this trend with their gradient-boosted convolutional neural network (GBCNN), where CNN layers captured non-linear solute–membrane interactions and the boosting stage provided interpretability through feature importance rankings. The GBCNN outperformed standalone CNN and RF models, confirming the added value of hybridisation.

Multimodal and hybrid DL approaches mark a shift from narrow, single-purpose models towards integrated frameworks capable of supporting both scientific discovery and operational optimisation. By fusing multiple data sources and combining architectures, these models can address the twin challenges of non-linear complexity and practical interpretability. The continued adoption of such approaches is likely to define the next generation of NF DL research, linking molecular-level descriptors with plant-scale operational intelligence.

5.5. Data scarcity and augmentation strategies

One of the key barriers to applying DL in NF is the limited dataset size. NF experiments are costly and time-consuming, leading to small and fragmented datasets. DL models, however, typically require large datasets to avoid overfitting. Zhang *et al.* (2021) addressed this problem by using data augmentation in their CNN-inspired representation learning model. By rationally distorting monomer structures and artificially expanding the dataset, they trained a DL model that achieved high accuracy

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

despite limited experimental data. This demonstrated that augmentation can partially compensate for data scarcity in NF. Although not yet fully explored in NF, some studies (Cao *et al.* 2024) have suggested that transfer learning pre-training DL models on large general datasets and fine-tuning them for NF applications could be an effective strategy to overcome small datasets.

5.6. Challenges of DL in NF

Although DL has shown remarkable promise for advancing NF research, its application remains constrained by several critical challenges. Perhaps the most significant limitation is data scarcity. Unlike fields such as computer vision or natural language processing, which benefit from massive datasets, NF studies are generally based on small experimental datasets due to the high cost, time, and resource demands of laboratory testing. This limitation is problematic because DL models, such as CNNs and RNNs, typically require large datasets to achieve generalisation and avoid overfitting. Data augmentation strategies can artificially expand available datasets and improve predictive performance, yet such methods cannot fully replicate the richness of diverse experimental data, leaving DL applications highly system-specific. A second challenge is the computational cost associated with training deep architectures. Unlike regression or tree-based models, which can be trained quickly on standard personal computers, CNNs and recurrent models demand significant computational resources, including high-performance GPUs. This requirement creates barriers for smaller laboratories or research groups without access to advanced computing facilities.

Another persistent concern is the black-box nature of DL. While DL models achieve high predictive accuracy, their internal decision-making processes are often difficult to interpret, limiting their acceptance in scientific and regulatory contexts. Although hybrid frameworks and explainability tools have been introduced to provide insights into variable importance and model reasoning, the broader challenge remains: DL predictions frequently lack transparent mechanistic grounding, which reduces their reliability for process design and optimisation. Finally, generalisation represents a substantial obstacle. Models trained on specific datasets often fail when applied to new feedwater chemistries, membrane materials, or operational conditions. This problem arises because DL models tend to capture dataset-specific correlations rather than universal mechanistic principles. Approaches such as transfer learning and digital twin frameworks hold promise in addressing this limitation, but practical demonstrations in NF remain scarce. ML models serving NF membrane applications.

Researchers have applied various ML models and algorithms for NF membrane applications, aiming at a wide range of purposes, including: (1) Operational performance evaluation and prediction, (2) Fabrication parameters optimisation, and (3) Fouling monitoring and cleaning schemes. The following sections illustrate these fields. Traditionally, most researchers tend to apply several ML algorithms/models then obtain and compare models' accuracy inferred by statistical parameters like regression R^2 and root mean square error (RMSE), towards selecting the most appropriate and accurate model. Other researchers would go further with an ensembled learning approach combining several models to produce a more accurate and robust result. The following sections illustrate the recent application of different ML algorithms/models in the three abovementioned fields focusing on field of application, algorithm used, and level of accuracy obtained.

6. NF MEMBRANE APPLICATIONS INTEGRATED WITH ML

6.1. Operational performance evaluation and prediction

From an operational point of view and for NF membrane filtering specific aqueous solution, NF membrane concerning parameters are flux determined by pressure magnitude, feed substances concentration determining rejection rate, and membrane recovery rate. These parameters are of interest to researchers to estimate and predict using various supervised ML algorithms, either using their own experimental data or by applying sets of data extracted from the literature. Practically, features of ML algorithms would be experimental data obtained for the operational parameters intended to be predicted, along with some additional influencing parameters, such as operational pressure, pH, solution temperature, and other relevant parameters.

Recently, ML prediction models have been widely applied for NF membranes performance evaluation (Ma *et al.* 2023; Zhu *et al.* 2023; Masuodi *et al.* 2024; Wang *et al.* 2024a; Sutariya *et al.* 2025). Ma *et al.* (2023) emphasised the relationship between structural and operational conditions of NF membrane applied for water desalination. They applied two ML algorithms: RF and XGBoost, to investigate the relationship between NF membrane structural parameters and operating conditions representing water/salt selectivity. Specifically, two key structural parameters of polyamide NF membranes (pore radius and zeta potential) and two operational parameters (applied pressure and feed concentration) were extracted from 100 journal publications (forming 360 data point) and analysed for their association with water/salt separation

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

performance. The models learned from these variables to evaluate their influence on selectivity. Predicted results revealed that structural parameters had a greater impact on water/salt selectivity than operating conditions, with the degree of influence varying according to salt type. For symmetrical salts, separation was predominantly governed by size sieving, whereas asymmetrical salts were mainly influenced by Donnan exclusion. Partial dependence plot analysis further elucidated structure/performance relationships between pore radius, zeta potential, and variations in water/salt selectivity. Masuodi *et al.* (2024) used an ML model to evaluate micropollutant removal efficiency of NF and RO membranes. In their study, they evaluated the predictive performance of three advanced ML models: ANN, RF, and XGBoost on micropollutant removal efficiency by NF and RO from water. Their investigation specifically explored the impact of MW and compound size influence on removal rates. Employing an extensive dataset, the ANN model was developed with a four-layer architecture comprising 128, 64, 32, and 32 nodes, trained over 2,500 epochs. The ANN model outperformed the other two models, achieving an (R^2: 0.9795), demonstrating its precision in capturing the variability inherent in RO and NF processes. The RF model (R^2: 0.9658) and the XGBoost model (R^2: 0.9447) also showed strong predictive capabilities, although they were slightly less accurate than the ANN. These findings highlight the potential of ML models, particularly ANNs, to optimise water substances removal techniques in RO and NF processes, providing valuable insights for enhancing the efficiency of these technologies. Based on NF membranes synthesised conditions, Sutariya *et al.* (2025) applied an ML-assisted performance prediction model to overcome the complexity of process permutation occurrence due to variables multitude involvement. According to their study, NF membranes key parameters (flux, rejection, and permeability) performance prediction required developing a conventional multiple linear regression and ML models (RF and multi-layer perceptron (MLP)). The input variables used to develop the predictive models include the concentration of piperazine (PIP) (0.01–2 wt%), trimesoylent chloride (TMC) (0.05–0.15 wt%), sodium lauryl sulfate (SLS) (0–10 mM), and reaction time (5–1,200 s). While the traditional regression model failed to accurately predict membrane performance, the RF model demonstrated superior predictive accuracy for permeability and NaCl rejection, achieving R^2 values of 0.9806 and 0.9812 for training, and 0.9669 and 0.9082 for testing, respectively. For Na_2SO_4 rejection, the MLP model outperformed the RF model, with R^2 values of 0.9972 for testing and 0.9844 for training. The optimal membrane performance was observed with permeance ranging from 16 to 20 LMH/bar and selectivity between NaCl and Na_2SO_4 exceeding 4,000. This performance was achieved under specific synthesis conditions: lower PIP concentrations (0.05–0.1 wt%), intermediate TMC concentrations (0.1 wt%), low SLS concentrations (1 mM), and shorter reaction times (5–30 s). The best-performing ML models provided valuable insights into the relative importance of these variables in determining membrane performance. This facilitates a deeper understanding of the correlation between model predictions and the fundamental principles of membrane fabrication processes, aiding in the design of high-performance NF membranes. Zhu *et al.* (2023) applied four traditional learning algorithms (MLR, SVM, ANN, kNN) and four ensemble learning algorithms (RF, GBDT, XGBoost, LightGBM) to predict contaminant removal efficiency in the NF membrane process. Their results indicate that ensemble models significantly outperformed traditional models in predictive accuracy. Additionally, an effective balance between accuracy and interpretability for the ensemble models through a robust model interpretation approach could be achieved. Feature selection was performed using the model interpretation method to identify the most influential variables affecting contaminant removal rates. Accordingly, the ensemble models retrained with the selected features demonstrated exceptional predictive performance, achieving adjusted R^2 values between 92.4 and 99.5%. Finally, the study revealed that MWCO of the membrane, McGowan volume of the solute (V), and MW of the compounds emerged as the most critical factors influencing contaminant removal in NF and RO processes. In general, ML-based models emerged as promising and further accurate approaches towards better performance evaluation and prediction modelling for NF membranes. Despite the strong predictive performance reported in these operational studies, several critical issues persist. Most models are trained on datasets compiled from either individual laboratories or literature sources that differ in analytical methods, membrane characterisation protocols, and reporting standards, which may introduce hidden biases. In addition, long-term stability, seasonal variability, and rare events are seldom represented in the training data, limiting the reliability of predictions for real plant operation. As a result, ML-based operational models for NF should currently be viewed as decision-support tools that complement, rather than replace, conventional engineering judgement and safety margins.

6.2. Fabrication parameters optimisation

Numerous techniques have been developed to enhance NF membrane performance. These methods aim to boost separation capabilities, raise productivity and selectivity, chemical resistance, improve thermal stability, lower rejection efficacy, and

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

progress in the membrane lifetime, and give antifouling features (Samavati *et al.* 2023; Liu *et al.* 2024). Among innovative fabrication, manufacturing, and modification procedures with a high fouling propensity and performance increase have been the subject of numerous research. This includes exploring fabrication and modification methods, including interfacial polymerisation (IP), grafting polymerisation (GP), hollow fibre, electrospun nanofibrous membranes (ENMs), phase inversion (PI), cross-linking (CL), layer-by-layer (LbL) technology, plasma polymerisation (PP), UV/photo grafting technique, plasma treatment, blending, and coating are some of the categories into which these techniques fall (An *et al.* 2023; Behroozi *et al.* 2023; Al Kholif *et al.* 2024). Practically, fabrication parameters are commonly including polymer type, monomer, catalyst concentrations, reaction time, and others (Bahreini *et al.* 2024). IP was applied by Usman *et al.* (2024) to fabricate three different NF membrane using three different cross-linkers for a combined application of saline water desalination and micropollutants removal. They used fabrication parameters including the concentration of PIP (0.01–2 wt%), the concentration of trimesoylchloride (TMC) (0.05–0.15 wt%), the concentration of SLS (0–10 mM), and reaction time (5–1,200 s) as input while having flux (before and after chlorination process) and rejection of sodium chloride and sodium sulphate as outputs. They explored the influence of different monomers on both structural and functional membranes features by applying Martin Gaussian Process Regression (MGPR) ML model, achieving as high as 99% accuracy. The database and information available for various types of polymers, especially those that have not been used or applied could be useful to train ML models towards selecting the best fit polymer for membrane fabrication. Wang *et al.* (2024a) introduced an ML and explainable-AI framework for predicting LBL NF membrane performance and identifying high-potential polymer candidates. Four ML models were evaluated, with XGBoost achieving the highest accuracy ($R^2 = 0.99$ for permeability, 0.80 for selectivity). Applying SHAP, their study revealed the influence of fabrication parameters and polymer structures, represented using Morgan fingerprints. Two reference fingerprints capturing beneficial atomic groups were used to screen 204 polymers from the Poly-info database, yielding 23 promising candidates for LBL NF membrane fabrication. OSN enables efficient separation of solvent–solute mixtures (200–1,000 Da) at mild temperatures without phase transitions, yet its industrial uptake is hindered by limited long-term stability data and a lack of predictive design tools (Gu *et al.* 2025). Goebel *et al.* (2020) extended an automated membrane-specific modelling framework, previously applied to solvent flux prediction, to quantify solute rejection in pure and mixed solvents. Using a PuraMem S600 membrane, they measured solute rejection across six solvents, and model candidates were automatically derived and validated. In their study, mixed solvent performance was further evaluated via membrane rejection maps using PuraMem 280 data. Their models achieved $<10\%$ deviation between measured and predicted rejection for most cases, with comparable accuracy in mixed solvents. Their approach offers a robust, data-driven tool for OSN process design, accelerating its deployment in chemical separations. It is worth noting that attempts to develop ML models based on NF membrane fabrication parameters are yet to be established or found in literature.

The reviewed work on fabrication optimisation underscores the promise of ML for accelerating membrane development; however, significant gaps remain. Most studies focus on relatively narrow regions of fabrication space and on a limited number of polymer systems, restricting the ability of models to extrapolate to genuinely novel chemistries or structures. Moreover, experimental datasets are rarely designed with ML in mind, leading to uneven coverage of key variables and difficulties in disentangling correlated effects. Until larger, systematically designed fabrication datasets become available for NF, ML-guided optimisation should be interpreted as a tool for hypothesis generation and local refinement rather than as a fully predictive design engine.

6.3. Fouling monitoring and cleaning schemes

The seriousness of the membrane fouling issue attracted the attention of many researchers to benefit from ML evolution towards studying and predicting concerning and influencing parameters in mitigation and cleaning processes. Practically, fouling is usually diagnosed by observing flux decline and selectivity deterioration represented by rejection reduction with time (Al Kholif *et al.* 2024; Fujioka *et al.* 2024). These parameters were targeted as feature inputs for ML algorithms in addition to other influencing parameters, including pH, temperature, and feedwater quality, while the outputs of such models included: recovery rate, permeate quality represented by TDS level, cake layer thickness, permeate flux, and fouling resistance (Wang *et al.* 2024c; Zheng *et al.* 2024b). Generally, identifying early indicators of fouling and optimising cleaning schedules, data-driven models like NN, SVM, and RF have been successfully used to analyse large volumes of process data (e.g., transmembrane pressure, permeate flux, conductivity, and chemical dosage) (Ignacz *et al.* 2025; Liang *et al.* 2025). Additionally, ML and AI-based systems can facilitate adaptive process control, which enhances membrane lifespan and lowers energy and chemical usage by automatically modifying operating parameters to reduce fouling in real time.

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

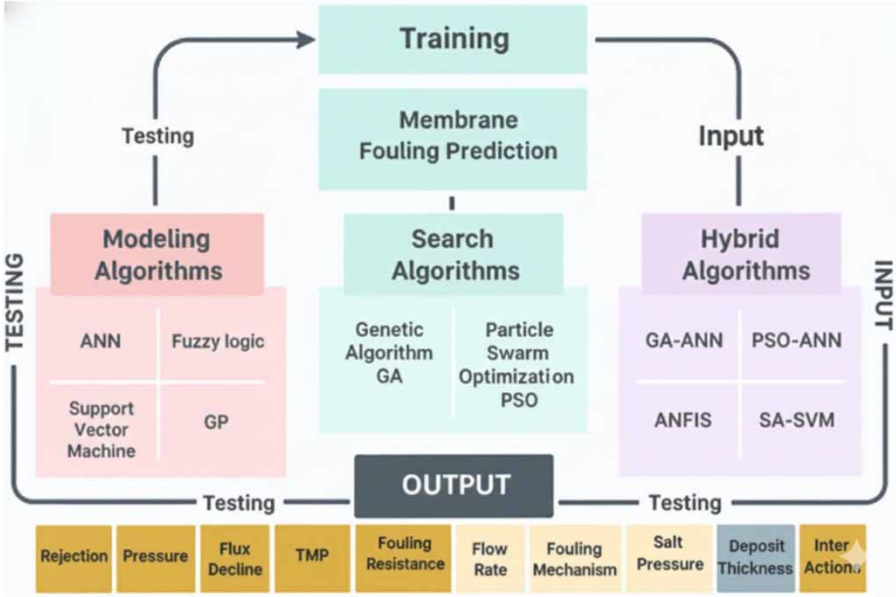


Figure 5 | Fouling prediction using ML (Habieeb *et al.* 2023).

Furthermore, to increase robustness and interpretability as well as closing the gap between theory and practice in NF system monitoring, hybrid models that combine physics-based techniques with ML algorithms are being created more frequently (Figure 5) (Habieeb *et al.* 2023). Smart, AI-assisted membrane systems are a major trend in next-generation water treatment technologies, as these advancements mark a change in NF operations from reactive to predictive maintenance.

Flux decline in NF/RO membrane processes, where membranes were fouled by synthesised humic acid, was investigated by Park *et al.* (2019). In their study, a CNN was developed to model membrane fouling in NF an RO using *in situ* optical coherence tomography (OCT) images. Trained on 13,708 high-resolution fouling layer images, the CNN simulated both organic fouling growth and flux decline, producing accurate 2D and 3D reconstructions. Compared with conventional mathematical models, their model achieved superior accuracy ($R^2 = 0.99$, RMSE = 2.82 μm for fouling growth; $R^2 = 0.99$, RMSE = 0.30 L m^{-2} h^{-1} for flux decline), demonstrating the potential of data-driven modelling as a robust alternative for predicting fouling and performance loss in high-pressure membrane systems. Technically, membrane fouling forecasting enables proactive strategies to sustain filtration performance. Shim *et al.* (2021) developed an LSTM model to predict permeate flux and fouling growth in NF system filtering NOM. Laboratory experiments using four NOMs foulants, namely, humic acid, bovine serum albumin, sodium alginate, and tannic acid, were conducted with real-time 2D OCT capturing cake layer formation. The LSTM, trained on experimental data, achieved RMSE below 1 L m^{-2} h^{-1} for flux and 10 μm for layer thickness. Their study demonstrates that obtaining high prediction accuracy was attributed to conducting the prediction for a mixture of NOM filtration rather than filtering a single solution which in turn positively affects the training information. Overall, ML-based fouling monitoring and cleaning optimisation frameworks provide clear advantages over purely empirical or rule-based approaches; nevertheless, their deployment faces several challenges. Many models are trained on relatively short experimental campaigns and do not account for long-term membrane ageing, changes in feedwater composition or sensor drift, all of which are common in practice. Furthermore, the lack of standardised performance indicators and benchmark datasets hampers objective comparison between competing approaches. To move from promising prototypes to robust operational tools, future studies will need to integrate ML with mechanistic fouling understanding, perform long-term validation in full-scale systems and explicitly consider the reliability and maintainability of sensing infrastructure.

6.4. RO ML modelling and algorithms serving NF

NF membranes can benefit from ML modelling and algorithms applied for RO in water applications, given the characteristics similarity between these membranes in terms of properties and performance features (Wang *et al.* 2024d). Mohammed *et al.* (2025) indicated that applying XGBoost outperformed other ML algorithms for the RO performance prediction in

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

desalination application. Such model is applicable for NF membranes, as the input parameters are typically observed for their performance evaluation, including: membrane's age, feed water temperature, pressure, feed water flow, and chloride concentration. The predictive performance found to be having high R^2 of 94.75 and low RMSE of 0.181, respectively. Operational features including feed flow rate, temperature, salt concentration, and pressure were used as inputs. A performance optimisation attempt that NF membrane desalination application may benefit from is by applying ANN models for performance optimisation. An optimised ANN by genetic algorithm optimised back-propagation neural network was applied by Wang *et al.* (2024d) for RO performance optimisation in brackish water desalination. In their study, they proposed several feed TDS scenarios with applying RSM as prior step to enhance ANN model performance forecasts NF membrane characterisation other than operational features can also benefit from some ML algorithms applied for RO membranes. Yeo *et al.* (2020) applied gradient boosting tree as a prediction model for thin-film composite RO membrane water permeability with loading, size, and pore size of the nanoparticle as inputs, confirming that great influence of these parameters on membrane performance and permeability.

7. DATA-DRIVEN MODELLING VS. MECHANISTIC AND CONVENTIONAL MODELLING IN NF

7.1. Advantages of ML models over conventional models

NF membranes have long been described by conventional mechanistic models, including the SDM, Spiegler–Kedem model, and extended Nernst–Planck equations. These models offer conceptual clarity and interpretability because they are based on thermodynamic (Bowen & Mohammad 1998; Geraldes & Afonso 2007). However, their accuracy is limited when applied to complicated or real-world systems since they frequently rely on simplifying assumptions (e.g., homogeneous membrane characteristics, steady-state circumstances, low concentration polarisation). Furthermore, mechanistic models usually necessitate an accurate understanding of system parameters, which can change over time due to fouling or aging and are challenging to assess *in situ*. Examples of these factors include membrane transport coefficients, solute diffusivity, and charge interactions. ML models, on the other hand, are excellent at extracting non-linear relationships and patterns from both historical and current data without the need for pre-existing physical assumptions (Figure 6). Even when mechanistic models are unsuccessful, methods such as ANNs, SVR, and ensemble learning can accurately forecast membrane performance (e.g., rejection rate, permeate flux) under a variety of circumstances (Masuodi *et al.* 2024; Sun *et al.* 2025). Additionally, ML models are particularly useful in dynamic operating situations because they can be updated with fresh data on a regular basis, allowing for adaptive learning and real-time decision assistance. ML models are a strong supplement or even a replacement for traditional modelling techniques in NF membrane applications because of their predictive power, data flexibility, and decreased reliance on system-specific constants, despite their potential lack of direct interpretability (Figure 6).

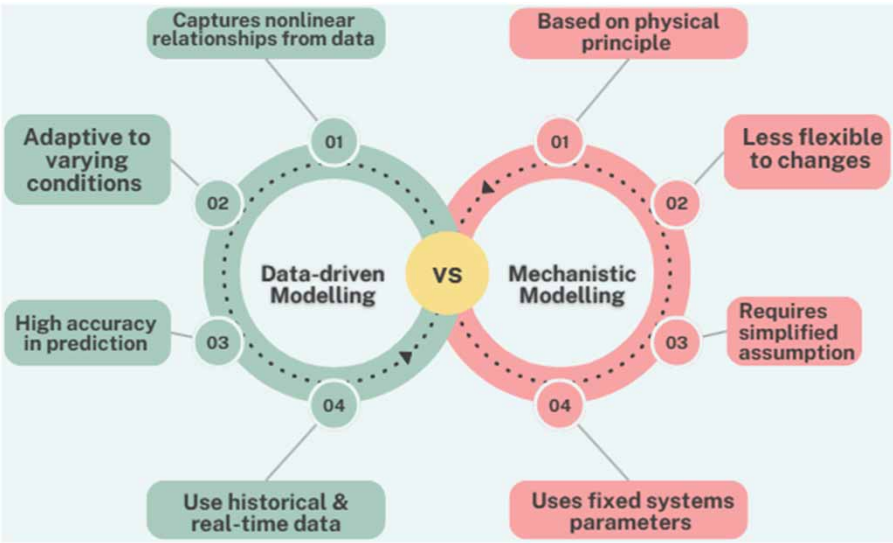


Figure 6 | Data-driven modelling vs. mechanistic modelling (prepared by the researchers).

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

7.2. Data quality and availability

Data quality and availability are critical factors in the success of a ML model, often having a far greater impact on performance than other aspects such as the choice of model architecture or the specific training procedures used. Despite this, much of the discourse outside the scientific community, especially in mainstream media, tends to focus on the development of cutting-edge ML models, such as the advent of transformers, the rapid growth of large language models, and other technological advancements. This is largely because discussions about the 'AI' itself are seen as more captivating than the more nuanced topic of the data that fuels these models. However, this can lead to the misconception that sophisticated models can compensate for poor data practices, which is far from the case. Several challenges related to data quality were pointed out in several studies, including issues with data integrity (Gudivada *et al.* 2017), concerns regarding the data lifecycle (Polyzotis *et al.* 2018), the integration with ML-OPS (Gudivada *et al.* 2017), and model management practices (Schelter *et al.* 2015). Additionally, some studies have proposed frameworks for assessing data quality (Gupta *et al.* 2021), automating data quality verification processes (Schelter *et al.* 2018), and methodologies for summarising dataset quality through tools like datasheets (Crawford 2021), nutritional labels (Stoyanovich & Howe 2019), and data cards (Tagliabue *et al.* 2021).

Historically, both academic research and industry efforts have concentrated on advancing or creating new ML models, often overlooking data challenges that extend beyond basic cleaning or pre-processing. However, the practical performance of AI-driven systems has demonstrated that it is ultimately constrained by the quality of the training data used (Polyzotis *et al.* 2019). Additionally, data typically have long lifecycles and are not confined to a single task; instead, they can be continuously utilised to refine and develop new models for solving emerging problems. This realisation has prompted a shift in research emphasis, moving from a model-centric to a data-centric approach in AI system development (Ré 2021). Contemporary AI applications demand vast amounts of training and testing data. This necessity presents significant challenges not only in ensuring the availability of sufficient data but also in maintaining its quality. For instance, incomplete, inaccurate, or unsuitable data can result in unreliable models that generate poor outcomes. Reliable AI systems depend on high-quality data across multiple dimensions, including accuracy, completeness, and consistency. Mohammed *et al.* (2025) examined the effect of six key data quality dimensions on ML performance across three distinct tasks: classification, regression, and clustering. Their study evaluated 19 commonly used ML algorithms on 10 diverse datasets, considering various scenarios of both high and low data quality through the constructing and investigating of three specific scenarios with three approaches: classification, regression, and clustering. The study revealed that data consistency and uniqueness had low effects while completeness and feature accuracy have high effects on the three scenarios accordingly.

8. FUTURE PROSPECTS AND CONCLUSIONS

The integration of ML and AI has established a new paradigm in NF membranes applications, moving predictive capabilities beyond the constraints of classical mechanistic modelling. As evidenced by the robust accuracy scored by ML modelling complex phenomena, ML is not merely a tool for pattern recognition; it is a catalyst for data-driven scientific discovery. However, translating such success into widespread industrial scalability and reliability requires acknowledging and systematically addressing persistent methodological bottlenecks. The following aspects are important to point out:

Technological implications and industrial scalability: the most significant technological implication of ML in NF is the realisation of proactive, adaptive process control. Future systems will integrate real-time sensor data with high accuracy predictive models (e.g., CNNs and LSTM) to create Digital Twins, enabling autonomous decision-making. This would involve:

- Autonomous optimisation: employing RL agents trained in simulated environments to continuously optimise operating parameters (pressure, crossflow velocity, feed concentration, and cleaning frequency) in real-time to maintain flux and maximise recovery, thereby reducing human intervention and energy costs.
- Predictive maintenance: utilising DL time-series models to predict fouling events days or weeks in advance, allowing for cleaning protocols to initiate proactively, thus extending membrane lifespan.
- Inverse design: leveraging generative ML models to identify the optimal chemical structures and process conditions for NF membranes tailored to specific source waters (e.g., high-salinity brine, complex industrial effluent), accelerating the design process and breaking the current permeability–selectivity trade-off.

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

Critical reflection of data, generalisation, and trust: the true limitations of the current ML-NF literature are not algorithmic but rather methodological, centering on data practices as follows:

- Data scarcity and heterogeneity: the reliance on small, single-laboratory datasets severely limits the generalisation of models. Future research must prioritise open science initiatives, creating large, standardised, multi-site datasets that capture operational, chemical, and structural heterogeneity, supported by unified data reporting protocols.
- Model generalisation (robustness): current models often capture system-specific correlations, failing when transferred to unseen facilities. To achieve industrial reliability, research must focus on physics-informed ML and hybrid ML-mechanistic frameworks. By embedding known transport equations (e.g., Nerst–Planck) into neural network loss functions, models can be constrained by physical laws, dramatically improving their robustness and external validity across different feedwater compositions.
- Industrial scalability and trust: Industrial adoption is hindered by the 'black-box' nature of complex models. Future efforts must mandate or encourage the use of explainable AI (XAI) as a standard practice, not an afterthought. XAI tools must be refined to translate statistical feature importance (e.g., SHAP values) directly, into clear mechanistic causality that researchers/engineers and regulators can trust, such as quantifying the exact contribution of the membrane's zeta potential versus the feed's pH on divalent ion rejection.

In summary, ML is set to revolutionise NF from laboratory synthesis to field operation. The technological implications point towards highly efficient, autonomous, and specialised NF systems. However, the researchers must prioritise solving the foundational problems of data standardisation and model transparency to ensure that these sophisticated tools successfully transition from promising research concepts to reliable, scalable solutions contributing meaningfully to global water treatment and security.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Az.A.: Writing – review & editing, writing – original draft, conceptualisation. Ah.A.: Writing – review & editing, and finalisation. H.I.: Writing – review & editing, writing – original draft, conceptualisation. L.Y.N.: Writing – review & editing, writing – original draft, vvisualisation, conceptualisation. S.H.: Writing – review & editing, writing – original draft, visualisation, conceptualisation. A.W.M.: Writing – review & editing, writing – original draft, supervision, conceptualisation.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used [ChatGPT] in order to paraphrase and improve some sentences. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

REFERENCES

Abba, S. I., Usman, J., Abdulazeez, I., Lawal, D. U., Baig, N., Usman, A. G. & Aljundi, I. H. (2023) Integrated modeling of hybrid nanofiltration/reverse osmosis desalination plant using deep learning-based crow search optimization algorithm, *Water*, **15** (19), 3515. https://doi.org/10.3390/W15193515.

Abdorrezaee, Z. & Raisi, A. (2021) A hybrid ultrafiltration/nanofiltration/pervaporation membrane process for intensifying the refining of crude canola oil and solvent recovery, *Chemical Engineering and Processing – Process Intensification*, **169**, 108598.

Abdu, F. J., Abba, S. I., Usman, J., Bala, A., Jibril, M. M., Shaik, F. & Aljundi, I. H. (2025) Design of real-time hybrid nanofiltration/reverse osmosis seawater desalination plant performance based on deep learning application, *Desalination*, **611**, 118918. https://doi.org/10.1016/J.DESAL.2025.118918.

Ahmed, D. F., Isawi, H., Badway, N. A., Elbayaa, A. A. & Shawky, H. (2021) Graphene oxide incorporated cellulose triacetate/cellulose acetate nanocomposite membranes for forward osmosis desalination, *Arabian Journal of Chemistry*, **14** (3), 102995. https://doi.org/10.1016/j.arabjc.2021.102995.

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

Al Kholif, M., Hermana, J., Bilad, M. R. & Chao, H. P. (2024) Comprehensive updates on recent advances, fouling mechanisms, and future perspectives of nanofiltration, *Journal of Water Process Engineering*, **63**, 105565. https://doi.org/10.1016/J.JWPE.2024.105565.

Alqadhi, N., Abdellah, M. H., Nematulloev, S., Mohammed, O. F., Abdulhamid, M. A. & Szekely, G. (2024) Solution-processable poly(ether-ether-ketone) membranes for organic solvent nanofiltration: from dye separation to pharmaceutical purification, *Separation and Purification Technology*, **328**, 125072. https://doi.org/10.1016/j.seppur.2023.125072.

An, M., Gutierrez, L., D'Haese, A., Tan, L., Verliefde, A. & Cornelissen, E. (2023) In-situ modification of nanofiltration and reverse osmosis membranes for organic micropollutants and salts removal: a review, *Desalination*, **565**, 116861. https://doi.org/10.1016/j.desal.2023.116861.

Bahreini, H., Ameri, E. & Ebadi-Dehaghani, H. (2024) Effect of incorporation of multi-walled carbon nanotubes on the CO_2/CH_4 separation performance of sulfonated poly (ether ether ketone)/polyetherimide composite membranes using design of experiments and molecular dynamics simulation methods, *Arabian Journal of Chemistry*, **17** (1), 105400.

Baig, N., Abba, S. I., Usman, J., Benaafi, M. & Aljundi, I. H. (2023a) Ensemble hybrid machine learning to simulate dye/divalent salt fractionation using a loose nanofiltration membrane, *Environmental Science: Advances*, **2** (10), 1446–1459. https://doi.org/10.1039/D3VA00124E.

Baig, N., Usman, J., Abba, S. I., Benaafi, M. & Aljundi, I. H. (2023b) Fractionation of dyes/salts using loose nanofiltration membranes: insight from machine learning prediction, *Journal of Cleaner Production*, **418**, 138193. https://doi.org/10.1016/J.JCLEPRO.2023.138193.

Behroozi, A. H. & Ataabadi, M. R. (2021) Improvement in microfiltration process of oily wastewater: a comprehensive review over two decades, *Journal of Environmental Chemical Engineering*, **9** (1), 104981. https://doi.org/10.1016/j.jece.2020.104981.

Behroozi, A. H., Al-Shaeli, M. & Vatanpour, V. (2023) Fabrication and modification of nanofiltration membranes by solution electrospinning technique: a review of influential factors and applications in water treatment, *Desalination*, **558**, 116638. https://doi.org/10.1016/j.desal.2023.116638.

Belhamidi, S., El-Ghzizel, S., Taky, M. & Elmidaoui, A. (2022) Nitrate removal of groundwater by reverse osmosis, nanofiltration and electrodialysis: performances and cost comparison, *Desalination and Water Treatment*, **262**, 338–346. https://doi.org/10.5004/dwt.2022.28602.

Bishop, C. M. & Nasrabadi, N. M. (2006) *Pattern Recognition and Machine Learning*, Vol. 4, Issue 4. London, UK: Springer.

Bonny, T., Kashkash, M. & Ahmed, F. (2022) An efficient deep reinforcement machine learning-based control reverse osmosis system for water desalination, *Desalination*, **522**, 115443. https://doi.org/10.1016/J.DESAL.2021.115443.

Borgomeo, E. (2024) A desalination research agenda fit for the 21st century, *Desalination*, **586**, 117910. https://doi.org/10.1016/j.desal.2024.117910.

Boussouga, Y.-A., Than, H. & Schäfer, A. I. (2022) Selenium species removal by nanofiltration: determination of retention mechanisms, *Science of The Total Environment*, **829**, 154287. https://doi.org/10.1016/j.scitotenv.2022.154287.

Bowen, W. R. & Mohammad, A. W. (1998) Characterization and prediction of nanofiltration membrane performance – a general assessment, *Chemical Engineering Research and Design*, **76** (8), 885–893. https://doi.org/10.1205/026387698525685.

Cao, Z., Barati Farimani, O., Ock, J. & Barati Farimani, A. (2024) Machine learning in membrane design: from property prediction to AI-guided optimization, *Nano Letters*, **24** (10), 2953–2960. https://doi.org/10.1021/ACS.NANOLETT.3C05137/ASSET/IMAGES/LARGE/NL3C05137_0004.JPEG.

Crawford, K. (2021) Datasheets for datasets, *Communications of the ACM*, **64** (12), 86–92.

Deen, W. M. (1987) Hindered transport of large molecules in liquid-filled pores, *AICHE Journal*, **33** (9), 1409–1425. https://doi.org/10.1002/aic.690330902.

Di, J., Wang, H., Zhang, L., Chen, Z., Zhang, Y., Mamba, B. B., Qiu, M., Guo, J. & Shao, L. (2024) Recent progress in advanced polyamide nanofiltration membranes via interfacial polymerization for desalination and beyond, *Desalination*, **592**, 118167. https://doi.org/10.1016/j.desal.2024.118167.

Donnan, F. G. (1995) Theory of membrane equilibria and membrane potentials in the presence of non-dialysing electrolytes. A contribution to physical-chemical physiology, *Journal of Membrane Science*, **100** (1), 45–55. https://doi.org/10.1016/0376-7388(94)00297-C.

Du, Y., Pramanik, B. K., Zhang, Y., Dumée, L. & Jegatheesan, V. (2022) Recent advances in the theory and application of nanofiltration: a review, *Current Pollution Reports*, **8** (1), 51–80. https://doi.org/10.1007/s40726-021-00208-1.

El-Aassar, A. M., Mahmoud, F. E., Elbakry, S., Sayed Alahl, A. A. & Isawi, H. (2024) Design, fabrication, and performance assessment for green hydrogen production unit, *International Journal of Hydrogen Energy*, **84**, 1050–1057. https://doi.org/10.1016/j.ijhydene.2024.08.197.

El-Ghzizel, S., Zeggar, H., Elmidaoui, A. & Taky, M. (2024) Nanofiltration powered by renewable energy for softening of slightly brackish-nitrated groundwater: sustainability study, *Journal of Water Process Engineering*, **60**, 105223. https://doi.org/10.1016/j.jwpe.2024.105223.

El-Noss, M., Isawi, H., Shawky, H. A., Gomaa, M. A. & Abdel-Mottaleb, M. S. A. (2020) Improvement of cellulose acetate forward osmosis membrane performance using zinc oxide nanoparticles, *Desalination and Water Treatment*, **193**, 19–33. https://doi.org/10.5004/dwt.2020.25677.

Fetanat, M., Keshtiara, M., Low, Z. X., Keyikoglu, R., Khataee, A., Orooji, Y., Chen, V., Leslie, G. & Razmjou, A. (2021) Machine learning for advanced design of nanocomposite ultrafiltration membranes, *Industrial and Engineering Chemistry Research*, **60** (14), 5236–5250. https://doi.org/10.1021/ACS.IECR.0C05446/ASSET/IMAGES/LARGE/IE0C05446_0010.JPEG.

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

Fujioka, T., Ung, C. T. L., Okuda, T. & Boivin, S. (2024) Controlling membrane fouling of nanofiltration using poly-aluminum chloride and *Moringa oleifera* coagulants, *Separation and Purification Technology*, **334**, 126016. https://doi.org/10.1016/j.seppur.2023.126016.

Gao, H., Zhong, S., Zhang, W., Igou, T., Berger, E., Reid, E., Zhao, Y., Lambeth, D., Gan, L., Afolabi, M. A., Tong, Z., Lan, G. & Chen, Y. (2022) Revolutionizing membrane design using machine learning-Bayesian optimization, *Environmental Science and Technology*, **56** (4), 2572–2581. https://doi.org/10.1021/ACS.EST.1C04373/SUPPL_FILE/ES1C04373_SI_002.XLSX.

Gao, H., Zhong, S., Dangayach, R. & Chen, Y. (2023) Understanding and designing a high-performance ultrafiltration membrane using machine learning, *Environmental Science and Technology*, **57** (46), 17831–17840. https://doi.org/10.1021/ACS.EST.2C05404/ASSET/IMAGES/LARGE/ES2C05404_0005.JPEG.

Geraldes, V. & Afonso, M. D. (2007) Prediction of the concentration polarization in the nanofiltration/reverse osmosis of dilute multi-ionic solutions, *Journal of Membrane Science*, **300** (1), 20–27. https://doi.org/10.1016/j.memsci.2007.04.025.

Giddings, J. C., Kucera, E., Russell, C. P. & Myers, M. N. (1968) Statistical theory for the equilibrium distribution of rigid molecules in inert porous networks. exclusion chromatography, *The Journal of Physical Chemistry*, **72** (13), 4397–4408. https://doi.org/10.1021/j100859a008.

Goebel, R., Glaser, T. & Skiborowski, M. (2020) Machine-based learning of predictive models in organic solvent nanofiltration: solute rejection in pure and mixed solvents, *Separation and Purification Technology*, **248**, 117046. https://doi.org/10.1016/J.SEPPUR.2020.117046.

Golabi, A., Erradi, A., Qiblawey, H., Tantawy, A., Bensaid, A. & Shaban, K. (2024) Optimal operation of reverse osmosis desalination process with deep reinforcement learning methods, *Applied Intelligence*, **54** (8), 6333–6353. https://doi.org/10.1007/S10489-024-05452-8.

Gong, W., Xu, H., Lu, J., Kim, J., Zhao, Y., Li, N., Zhang, Y., Yang, J., Xu, D. & Liang, H. (2024) Gradient boosting decision tree algorithms for accelerating nanofiltration membrane design and discovery, *Desalination*, **592**, 118072. https://doi.org/10.1016/J.DESAL.2024.118072.

Gu, S., Li, S. & Xu, Z. (2025) Organic solvent nanofiltration membranes for separation in non-polar solvent system, *Green Energy & Environment*, **10** (2), 244–267. https://doi.org/10.1016/j.gee.2024.02.007.

Gudivada, V., Apon, A. & Ding, J. (2017) Data quality considerations for big data and machine learning: going beyond data cleaning and transformations, *International Journal on Advances in Software*, **10** (1), 1–20.

Gupta, N., Patel, H., Afzal, S., Panwar, N., Mittal, R. S., Guttula, S., Jain, A., Nagalapatti, L., Mehta, S. & Hans, S. (2021) Data Quality Toolkit: automatic assessment of data quality and remediation for machine learning datasets. *ArXiv Preprint ArXiv:2108.05935*.

Habieeb, A. R., Kabeel, A. E., Sultan, G. I. & Abdelsalam, M. M. (2023) Advancements in water desalination through artificial intelligence: a comprehensive review of AI-based methods for reverse osmosis membrane processes, *Water Conservation Science and Engineering*, **8** (1), 53. https://doi.org/10.1007/s41101-023-00227-7.

Hosseinzadeh, A., Zhou, J. L., Zyaie, J., AlZainati, N., Ibrar, I. & Altaee, A. (2022) Machine learning-based modeling and analysis of PFOS removal from contaminated water by nanofiltration process, *Separation and Purification Technology*, **289**, 120775. https://doi.org/10.1016/J.SEPPUR.2022.120775.

Hu, J., Kim, C., Halasz, P., Kim, J. F., Kim, J. & Szekely, G. (2021) Artificial intelligence for performance prediction of organic solvent nanofiltration membranes, *Journal of Membrane Science*, **619**, 118513. https://doi.org/10.1016/J.MEMSCI.2020.118513.

Hu, A., Liu, Y., Wang, X., Xia, S. & Van der Bruggen, B. (2025) A machine learning based framework to tailor properties of nanofiltration and reverse osmosis membranes for targeted removal of organic micropollutants, *Water Research*, **268**, 122677. https://doi.org/10.1016/J.WATRES.2024.122677.

Idrissi, Y. E. L., Benabbou, M., Rais, Z. & Haji, M. E. L. (2024) Brackish and seawater pretreatment processes: a systematic literature review, *Desalination and Water Treatment*, **318**, 100350.

Ignacz, G., Alqadhi, N. & Szekely, G. (2023a) Explainable machine learning for unraveling solvent effects in polyimide organic solvent nanofiltration membranes, *Advanced Membranes*, **3**, 100061. https://doi.org/10.1016/J.ADVMEM.2023.100061.

Ignacz, G., Beke, A. K. & Szekely, G. (2023b) Data-driven future for nanofiltration: escaping linearity, *Journal of Membrane Science Letters*, **3** (1), 100040. https://doi.org/10.1016/J.MEMLET.2023.100040.

Ignacz, G., Bader, L., Beke, A. K., Ghunaim, Y., Shastry, T., Vovusha, H., Carbone, M. R., Ghanem, B. & Szekely, G. (2025) Machine learning for the advancement of membrane science and technology: a critical review, *Journal of Membrane Science*, **713**, 123256. https://doi.org/10.1016/J.MEMSCI.2024.123256.

Isawi, H. (2018) Development of thin-film composite membranes via radical grafting with methacrylic acid/ZnO doped TiO_2 nanocomposites, *Reactive and Functional Polymers*, **131**, 400–413. https://doi.org/10.1016/j.reactfunctpolym.2018.08.018.

Isawi, H. (2019) Evaluating the performance of different nano-enhanced ultrafiltration membranes for the removal of organic pollutants from wastewater, *Journal of Water Process Engineering*, **31**, 100833. https://doi.org/10.1016/j.jwpe.2019.100833.

Isawi, H. (2023) Synthesis of graphene oxide-silver (GO-Ag) nanocomposite TFC RO membrane to enhance morphology and separation performances for groundwater desalination (case study Marsa Alam area – Red Sea), *Chemical Engineering and Processing – Process Intensification*, **187**, 109343. https://doi.org/10.1016/j.cep.2023.109343.

Isawi, H. (2024) Using zero nano-valent iron/thin film composite (ZNVI/TFC) membrane for brackish water desalination and purification, *Surfaces and Interfaces*, **54**, 105237. https://doi.org/10.1016/j.surfin.2024.105237.

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

Isawi, H., El-Sayed, M. H., Feng, X., Shawky, H. & Abdel Mottaleb, M. S. (2016) Surface nanostructuring of thin film composite membranes via grafting polymerization and incorporation of ZnO nanoparticles, *Applied Surface Science*, **385**, 268–281. https://doi.org/10.1016/j.apsusc.2016.05.141.

Jafari, M., Tzirtzipi, C., Aghdam, A. M., Chahartagh, N. M. & Dominguez, B. C. (2025) AI-assisted prediction and optimization of micropollutants removal with forward osmosis membranes, *Journal of Membrane Science*, **733**, 124346. https://doi.org/10.1016/J.MEMSCI.2025.124346.

Jeong, N., Chung, T. H. & Tong, T. (2021) Predicting micropollutant removal by reverse osmosis and nanofiltration membranes: is machine learning viable? *Environmental Science and Technology*, **55** (16), 11348–11359. https://doi.org/10.1021/ACS.EST.1C04041/SUPPL_FILE/ES1C04041_SI_003.XLSX.

Jeong, H., Yun, B., Na, S., Son, M., Chae, S. H., Kim, C. M. & Cho, K. H. (2024) Multimodal deep learning models incorporating the adsorption characteristics of the adsorbent for estimating the permeate flux in dynamic membranes, *Journal of Membrane Science*, **709**, 123105. https://doi.org/10.1016/J.MEMSCI.2024.123105.

Ji, Z., Guan, H. & Dong, Y. (2025) Performance and mechanism of lithium extraction from water via machine learning-powered nanofiltration, *Journal of Membrane Science*, **733**, 124344. https://doi.org/10.1016/J.MEMSCI.2025.124344.

Laurell, P., Sivonen, K., Hesampour, M., Tuutijärvi, T. & Vahala, R. (2024) Application and environmental impact of loose nanofiltration in surface water treatment, *Journal of Water Process Engineering*, **60**, 105174. https://doi.org/10.1016/j.jwpe.2024.105174.

Lee, S., Shirts, M. R. & Straub, A. P. (2024) Molecular fingerprint-aided prediction of organic solute rejection in reverse osmosis and nanofiltration, *Journal of Membrane Science*, **705**, 122927. https://doi.org/10.1016/J.MEMSCI.2024.122927.

Levenstein, R., Hasson, D. & Semiat, R. (1996) Utilization of the Donnan effect for improving electrolyte separation with nanofiltration membranes, *Journal of Membrane Science*, **116** (1), 77–92. https://doi.org/10.1016/0376-7388(96)00029-4.

Li, J., Xu, C., Long, J., Ding, Z., Yuan, R. & Li, Z. (2022) Lamellar MXene nanofiltration membranes for electrostatic modulation of molecular permeation: implications for fine separation, *ACS Applied Nano Materials*, **5** (5), 7373–7381.

Li, H., Zeng, B., Qiu, T., Huang, W., Wang, Y., Sheng, G. P. & Wang, Y. (2023) Deep learning models for assisted decision-making in performance optimization of thin film nanocomposite membranes, *Journal of Membrane Science*, **687**, 122093. https://doi.org/10.1016/j.memsci.2023.122093.

Li, H., Zeng, B., Tuo, J., Wang, Y., Sheng, G. P. & Wang, Y. (2024) Development of an improved deep network model as a general technique for thin film nanocomposite reverse osmosis membrane simulation, *Journal of Membrane Science*, **692**, 122320. https://doi.org/10.1016/J.MEMSCI.2023.122320.

Liang, L., Lu, D., Qin, Y., Chen, J. & Zhang, L. (2025) Machine learning in membrane science: bridging materials, structures, and performance for next-generation membrane design, *Separation and Purification Technology*, **369**, 133091. https://doi.org/10.1016/J.SEPPUR.2025.133091.

Liu, T., Liu, L., Cui, F., Ding, F., Zhang, Q. & Li, Y. (2020) Predicting the performance of polyvinylidene fluoride, polyethersulfone and polysulfone filtration membranes using machine learning, *Journal of Materials Chemistry A*, **8** (41), 21862–21871. https://doi.org/10.1039/D0TA07607D.

Liu, Y., Ran, J., Guo, Q., Gao, L., Yin, S., Li, S., Jiang, X., Zhang, N. & Zhou, G. (2024) Advancements in nanofiltration fouling phenomenon: from water treatment to salt lakes environments, *Desalination*, **583**, 117649. https://doi.org/10.1016/j.desal.2024.117649.

Liu, P., Xu, H., Jin, P., Zhu, X., Zheng, J., Liu, Y., Yang, J., Xu, D. & Liang, H. (2025) DFT-assisted machine learning for polyester membrane design in textile wastewater recovery applications, *Water Research*, **279**, 123438. https://doi.org/10.1016/J.WATRES.2025.123438.

Lu, D., Ma, X., Lu, J., Qian, Y., Geng, Y., Wang, J., Yao, Z., Liang, L., Sun, Z., Liang, S. & Zhang, L. (2023) Ensemble machine learning reveals key structural and operational features governing ion selectivity of polyamide nanofiltration membranes, *Desalination*, **564**, 116748. https://doi.org/10.1016/J.DESAL.2023.116748.

Lu, Q., Zhang, H., Fan, R., Wan, Y. & Luo, J. (2025) Machine learning-based Bayesian optimization facilitates ultrafiltration process design for efficient protein purification, *Separation and Purification Technology*, **363**, 132122. https://doi.org/10.1016/J.SEPPUR.2025.132122.

Ma, X., Lu, D., Lu, J., Qian, Y., Zhang, S., Yao, Z., Liang, L., Sun, Z. & Zhang, L. (2023) Revealing key structural and operating features on water/salts selectivity of polyamide nanofiltration membranes by ensemble machine learning, *Desalination*, **548**, 116293. https://doi.org/10.1016/J.DESAL.2022.116293.

Madhavan, G. (2025) *Mastering Machine Learning: From Basics to Advanced*. India: Springer.

Mallya, D. S., Abdikheibari, S., Dumée, L. F., Muthukumaran, S., Lei, W. & Baskaran, K. (2023) Removal of natural organic matter from surface water sources by nanofiltration and surface engineering membranes for fouling mitigation – a review, *Chemosphere*, **321**, 138070. https://doi.org/10.1016/j.chemosphere.2023.138070.

Masuodi, P., Bahmanzadegan, F., Hemmati, A. & Ghaemi, A. (2024) Evaluating the efficiency of nanofiltration and reverse osmosis membranes for the removal of micro-pollutants using a machine learning approach, *Case Studies in Chemical and Environmental Engineering*, **9**, 100750. https://doi.org/10.1016/J.CSCEE.2024.100750.

Meng, Z., Zhu, L., Wang, X. & Zhu, M. (2023) Electrospun nanofibrous composite membranes for separations, *Accounts of Materials Research*, **4** (2), 180–192. https://doi.org/10.1021/accountsmr.2c00219.

Mitchell, T. (1997) *Machine Learning*. New York: McGraw-Hill Higher Education.

Mohammad, A. W., Teow, Y. H., Ang, W. L., Chung, Y. T., Oatley-Radcliffe, D. L. & Hilal, N. (2015) Nanofiltration membranes review: recent advances and future prospects, *Desalination*, **356**, 226–254. https://doi.org/10.1016/j.desal.2014.10.043.

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

Mohammed, S., Budach, L., Feuerpfeil, M., Ihde, N., Nathansen, A., Noack, N., Patzlaff, H., Naumann, F. & Harmouch, H. (2025) The effects of data quality on machine learning performance on tabular data, *Information Systems*, **132**, 102549. https://doi.org/10.1016/j.is.2025.102549.

Niu, Y., Shi, H., Zheng, W., Chen, Y., Li, T. & Dong, B. (2025) A study of the effect of dissolved organic matter compositions on membrane fouling prediction in combined nanofiltration processes, *Separation and Purification Technology*, **378**, 134763. https://doi.org/10.1016/J.SEPPUR.2025.134763.

Park, S., Baek, S.-S., Pyo, J., Pachepsky, Y., Park, J. & Cho, K. H. (2019) Deep neural networks for modeling fouling growth and flux decline during NF/RO membrane filtration, *Journal of Membrane Science*, **587**, 117164. https://doi.org/10.1016/j.memsci.2019.06.004.

Park, S., Shim, J., Yoon, N., Lee, S., Kwak, D., Lee, S., Kim, Y. M., Son, M. & Cho, K. H. (2022) Deep reinforcement learning in an ultrafiltration system: optimizing operating pressure and chemical cleaning conditions, *Chemosphere*, **308**, 136364. https://doi.org/10.1016/J.CHEMOSPHERE.2022.136364.

Parmar, V. P. & Dhruv, A. J. (2021) 'Efficient sea water purification using hybrid nanofiltration system and ML for optimization', *Proceedings – 2021 1st IEEE International Conference on Artificial Intelligence and Machine Vision, AIMV 2021*.

Petukhov, D. I. & Johnson, D. J. (2024) Membrane modification with carbon nanomaterials for fouling mitigation: a review, *Advances in Colloid and Interface Science*, **327**, 103140. https://doi.org/10.1016/j.cis.2024.103140.

Polyzotis, N., Roy, S., Whang, S. E. & Zinkevich, M. (2018) Data lifecycle challenges in production machine learning: a survey, *ACM Sigmod Record*, **47** (2), 17–28.

Polyzotis, N., Zinkevich, M., Roy, S., Breck, E. & Whang, S. (2019) Data validation for machine learning, *Proceedings of Machine Learning and Systems*, **1**, 334–347.

Qrenawi, L. I. & Abuhabib, A. A. (2020) A review on sources, types, mechanisms, characteristics, impacts and control strategies of fouling in RO membrane systems, *Desalination and Water Treatment*, **208**, 43–69. https://doi.org/10.5004/dwt.2020.26448.

Ré, C. (2021) *The Road to Software 2.0 or Data-Centric AI* [Distinguished Lecture]. Department of Computer Science, University of Toronto, Toronto, Canada. https://cs.toronto.edu/news-events/events/distinguished-lecture-series-chris-re/.

Roselli, M., Onesti, R., Boi, C. & Bandini, S. (2025) Recovery of lactose from acid whey by nanofiltration: an experimental study, *Separation and Purification Technology*, **353**, 128303.

Saeid Rostami, M. & Mehdi Khodaei, M. (2023) Investigation of permeability, hydrophilicity, and pollutant removal properties of PES/MIL-53(Al)@MWCNT nanofiltration membrane, *Chemical Engineering Journal*, **475**, 146074. https://doi.org/10.1016/j.cej.2023.146074.

Samavati, Z., Samavati, A., Goh, P. S., Fauzi Ismail, A. & Sohaimi Abdullah, M. (2023) A comprehensive review of recent advances in nanofiltration membranes for heavy metal removal from wastewater, *Chemical Engineering Research and Design*, **189**, 530–571. https://doi.org/10.1016/j.cherd.2022.11.042.

Samavati, Z., Samavati, A., Goh, P. S., Fauzi Ismail, A., Yusof, N., Sohaimi Abdullah, M., Hashim, N., Kerisnan, N. D. A. P., Yahaya, N. K. E. M. & Idris, A. (2024) Recent advances in modifying the surface of polymeric NF membranes to enhance the removal of endocrine-disrupting compounds from water and wastewater, *Journal of Environmental Chemical Engineering*, **12** (1), 111696. https://doi.org/10.1016/j.jece.2023.111696.

Schelter, S., Biessmann, F., Januschowski, T., Salinas, D., Seufert, S. & Szarvas, G. (2015) *On Challenges in Machine Learning Model Management*. Berlin, Germany: IEEE Data Engineering Bulletin.

Schelter, S., Lange, D., Schmidt, P., Celikel, M., Biessmann, F. & Grafberger, A. (2018) Automating large-scale data quality verification, *Proceedings of the VLDB Endowment*, **11** (12), 1781–1794.

Shahlol, O. M. A., Isawi, H., El-Malky, M. G., Al-Aassar, A. E.-H. M. & Zwai, A. E. (2020) Performance evaluation of the different nano-enhanced polysulfone membranes via membrane distillation for produced water desalination in Sert Basin-Libya, *Arabian Journal of Chemistry*, **13** (4), 5118–5136. https://doi.org/10.1016/j.arabjc.2020.02.011.

Shahouni, R., Abbasi, M., Dibaj, M. & Akrami, M. (2024) Utilising artificial intelligence to predict membrane behaviour in water purification and desalination, *Water*, **16** (20), 2940. https://doi.org/10.3390/W16202940.

Shim, J., Park, S. & Cho, K. H. (2021) Deep learning model for simulating influence of natural organic matter in nanofiltration, *Water Research*, **197**, 117070. https://doi.org/10.1016/J.WATRES.2021.117070.

Singh, S., Suthar, G., Bhushan Gupta, A. & Bezbaruah, A. N. (2025) Machine learning approach for predicting perfluorooctanesulfonate rejection in efficient nanofiltration treatment and removal, *ACS ES&T Water*, **5** (3), 1216–1228. https://doi.org/10.1021/ACSESTWATER.4C01003/ASSET/IMAGES/LARGE/EW4C01003_0005.JPEG.

Stoyanovich, J. & Howe, B. (2019) Nutritional labels for data and models, *Quarterly Bulletin of the Computer Society of the IEEE Technical Committee on Data Engineering*, **42** (3), 13–23.

Sun, J. O., Hua, T. W., Guan, Y. F. & Yu, H. Q. (2025) Predicting and understanding the performance of polyamide nanofiltration membrane for Li/Mg selective separation based on machine learning, *Water Research*, **285**, 124140. https://doi.org/10.1016/J.WATRES.2025.124140.

Sutariya, B., Sarkar, P., Indurkar, P. D. & Karan, S. (2025) Machine learning-assisted performance prediction from the synthesis conditions of nanofiltration membranes, *Separation and Purification Technology*, **354**, 128960. https://doi.org/10.1016/J.SEPPUR.2024.128960.

Tagliabue, J., Tuulos, V., Greco, C. & Dave, V. (2021) DAG card is the new model card. *ArXiv Preprint ArXiv:2110.13601*.

Tsuru, T., Nakao, S. & Kimura, S. (1991) Calculation of ion rejection by extended Nernst–Planck equation with charged reverse osmosis membranes for single and mixed electrolyte solutions, *Journal of Chemical Engineering of Japan*, **24** (4), 511–517.

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

Usha, Z. R., Liu, C., Zhang, S. & Wang, Z. (2025) A comprehensive study of recent advances, challenges and factors of nanofiltration membrane in various industrial applications, *Desalination*, **599**, 118461. https://doi.org/10.1016/j.desal.2024.118461.

Usman, J., Baig, U., Abba, S. I., Alharthi, F. A., Fellows, C. M., Waheed, A. & Aljundi, I. H. (2024) Tailoring thin film composite membranes for clean water production: a study on structural variations and predictive insights using machine learning, *Journal of Environmental Chemical Engineering*, **12** (3), 112569. https://doi.org/10.1016/j.jece.2024.112569.

Voutchkov, N. (2018) Energy use for membrane seawater desalination – current status and trends, *Desalination*, **431**, 2–14.

Waheed, A., Baig, U. & Aljundi, I. H. (2024) Design and fabrication of thin film composite organic solvent nanofiltration membranes by making use of the chirality of *trans*-1,2-diaminocyclohexane; characterization and performance studies, *Colloid and Interface Science Communications*, **61**, 100790. https://doi.org/10.1016/j.colcom.2024.100790.

Wang, L., Rehman, D., Sun, P.-F., Deshmukh, A., Zhang, L., Han, Q., Yang, Z., Wang, Z., Park, H.-D., Lienhard, J. H. & Tang, C. Y. (2021) Novel positively charged metal-coordinated nanofiltration membrane for lithium recovery, *ACS Applied Materials & Interfaces*, **13** (14), 16906–16915.

Wang, C., Wang, L., Soo, A., Bansidhar Pathak, N. & Kyong Shon, H. (2023a) Machine learning based prediction and optimization of thin film nanocomposite membranes for organic solvent nanofiltration, *Separation and Purification Technology*, **304**, 122328. https://doi.org/10.1016/J.SEPPUR.2022.122328.

Wang, F., Wang, W., Wang, H., Zhao, Z., Zhou, T., Jiang, C., Li, J., Zhang, X., Liang, T. & Dong, W. (2023b) Experiments and machine learning-based modeling for haloacetic acids rejection by nanofiltration: influence of solute properties and operating conditions, *Science of The Total Environment*, **883**, 163610. https://doi.org/10.1016/J.SCITOTENV.2023.163610.

Wang, M., Shi, G. M., Zhao, D., Liu, X. & Jiang, J. (2023c) Machine learning-assisted design of thin-Film composite membranes for solvent recovery, *Environmental Science and Technology*, **57** (42), 15914–15924. https://doi.org/10.1021/ACS.EST.3C04773/ASSET/IMAGES/LARGE/ES3C04773_0006.JPEG.

Wang, C., Wang, L., Yu, H., Seo, A., Wang, Z., Rajabzadeh, S., Ni, B. J. & Shon, H. K. (2024a) Machine learning for layer-by-layer nanofiltration membrane performance prediction and polymer candidate exploration, *Chemosphere*, **350**, 140999. https://doi.org/10.1016/J.CHEMOSPHERE.2023.140999.

Wang, H., Zeng, J., Dai, R. & Wang, Z. (2024b) Understanding rejection mechanisms of trace organic contaminants by polyamide membranes via data-knowledge codriven machine learning, *Environmental Science and Technology*, **58** (13), 5878–5888. https://doi.org/10.1021/ACS.EST.3C08523/ASSET/IMAGES/LARGE/ES3C08523_0006.JPEG.

Wang, L., Li, Z., Fan, J. & Han, Z. (2024c) The intelligent prediction of membrane fouling during membrane filtration by mathematical models and artificial intelligence models, *Chemosphere*, **349**, 141031. https://doi.org/10.1016/j.chemosphere.2023.141031.

Wang, X., Sun, X., Wu, Y., Gao, F. & Yang, Y. (2024d) Optimizing reverse osmosis desalination from brackish waters: predictive approach employing response surface methodology and artificial neural network models, *Journal of Membrane Science*, **704**, 122883. https://doi.org/10.1016/J.MEMSCI.2024.122883.

Wang, Y., Yan, H., Ma, M., Zhou, C., Hou, Q., Xu, Z., Xia, M., Ni, H., Jia, Y. & Ye, P. (2024e) Graft-crosslinked graphene oxide nanofiltration membrane with pH-responsive nanochannels, *Polymers for Advanced Technologies*, **35** (1), e6250.

Wang, C., Jiao, F., Zhang, S., Gou, R., Li, H., Liu, Y. & Zhou, X. (2025) Machine learning-enhanced morphology and size control of 1,3,5-trinitro-1,3,5-triazinane (RDX) in nanofiltration-assisted cooling crystallization, *Separation and Purification Technology*, **374**, 133557. https://doi.org/10.1016/J.SEPPUR.2025.133557.

Xiao, J., Li, H., Hu, J., Wang, Y. & Wang, Y. (2025) Optimizing nanofiltration for PFOS removal: a gradient boosting convolutional model approach, *Journal of Environmental Chemical Engineering*, **13** (4), 117384. https://doi.org/10.1016/J.JECE.2025.117384.

Xu, R., Zhang, Z., Deng, C., Nie, C., Wang, L., Shi, W., Lyu, T. & Yang, Q. (2024) Micropollutant rejection by nanofiltration membranes: a mini review dedicated to the critical factors and modelling prediction, *Environmental Research*, **244**, 117935. https://doi.org/10.1016/j.envres.2023.117935.

Yeo, C. S. H., Xie, Q., Wang, X. & Zhang, S. (2020) Understanding and optimization of thin film nanocomposite membranes for reverse osmosis with machine learning, *Journal of Membrane Science*, **606**, 118135. https://doi.org/10.1016/J.MEMSCI.2020.118135.

Yogarathinam, L. T., Abba, S. I., Usman, J., Lawal, D. U. & Aljundi, I. H. (2024) Predicting micropollutant removal through nanopore-sized membranes using several machine-learning approaches based on feature engineering, *RSC Advances*, **14** (27), 19331–19348. https://doi.org/10.1039/D4RA02475C.

Yoon, N., Park, S., Son, M. & Cho, K. H. (2022) Automation of membrane capacitive deionization process using reinforcement learning, *Water Research*, **227**, 119337. https://doi.org/10.1016/J.WATRES.2022.119337.

Zahid, M., Rashid, A., Akram, S., Rehan, Z. A. & Razzaq, W. (2018) A comprehensive review on polymeric nano-composite membranes for water treatment, *Journal of Membrane Science & Technology*, **8** (1), 1–20.

Zhang, Z., Luo, Y., Peng, H., Chen, Y., Liao, R. Z. & Zhao, Q. (2021) Deep spatial representation learning of polyamide nanofiltration membranes, *Journal of Membrane Science*, **620**, 118910. https://doi.org/10.1016/J.MEMSCI.2020.118910.

Zhang, B. W., Xu, X., Zhang, Z. Z. & Yao, L. (2024) 'Prediction and modeling of desalination performance of nanofiltration membranes based on machine learning', *2024 3rd International Conference on Artificial Intelligence and Computer Information Technology, AICIT 2024*.

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest

Zhang, Y., Han, B., Song, Q., Jin, S., Lu, Z., Lu, S., Zhang, X., Zhang, Y. & Li, D. (2025) Revealing key structural and operating parameters on salt/dye separation of loose nanofiltration membrane by ensemble machine learning, *Journal of Membrane Science*, **732**, 124274. https://doi.org/10.1016/J.MEMSCI.2025.124274.

Zheng, R., Xu, S., Zhong, S., Tong, X., Yu, X., Zhao, Y. & Chen, Y. (2024a) Enhancing ion selectivity of nanofiltration membranes via heterogeneous charge distribution, *Environmental Science and Technology*, **58** (51), 22818–22828. https://doi.org/10.1021/ACS.EST.4C08841/ASSET/IMAGES/LARGE/ES4C08841_0004.JPEG.

Zheng, W., Chen, Y., Xu, X., Peng, X., Niu, Y., Xu, P. & Li, T. (2024b) Research on the factors influencing nanofiltration membrane fouling and the prediction of membrane fouling, *Journal of Water Process Engineering*, **59**, 104876. https://doi.org/10.1016/J.JWPE.2024.104876.

Zhu, T., Zhang, Y., Tao, C., Chen, W. & Cheng, H. (2023) Prediction of organic contaminant rejection by nanofiltration and reverse osmosis membranes using interpretable machine learning models, *Science of The Total Environment*, **857**, 159348. https://doi.org/10.1016/J.SCITOTENV.2022.159348.

Zhu, H., Szymczyk, A. & Ghoufi, A. (2024) Multiscale modelling of transport in polymer-based reverse-osmosis/nanofiltration membranes: present and future, *Discover Nano*, **19** (1), 1–21. https://doi.org/10.1186/S11671-024-04020-W.

First received 7 October 2025; accepted in revised form 13 January 2026. Available online 25 February 2026

Downloaded from iwaponline.com/wst/article-pdf/93/6/817/1611977/wst2026234.pdf
by guest